

Reddit India — Bot Activity Report

Comprehensive Statistical Analysis

25 Subreddits · 13 Months · 9,971 Posts · 31,204 Comment Rows

July 22, 2026

§1 Data Portrait

§2 Univariate Analysis

§3 Signal Correlations

§4 Text Intelligence

§5 Cross-Sub Network

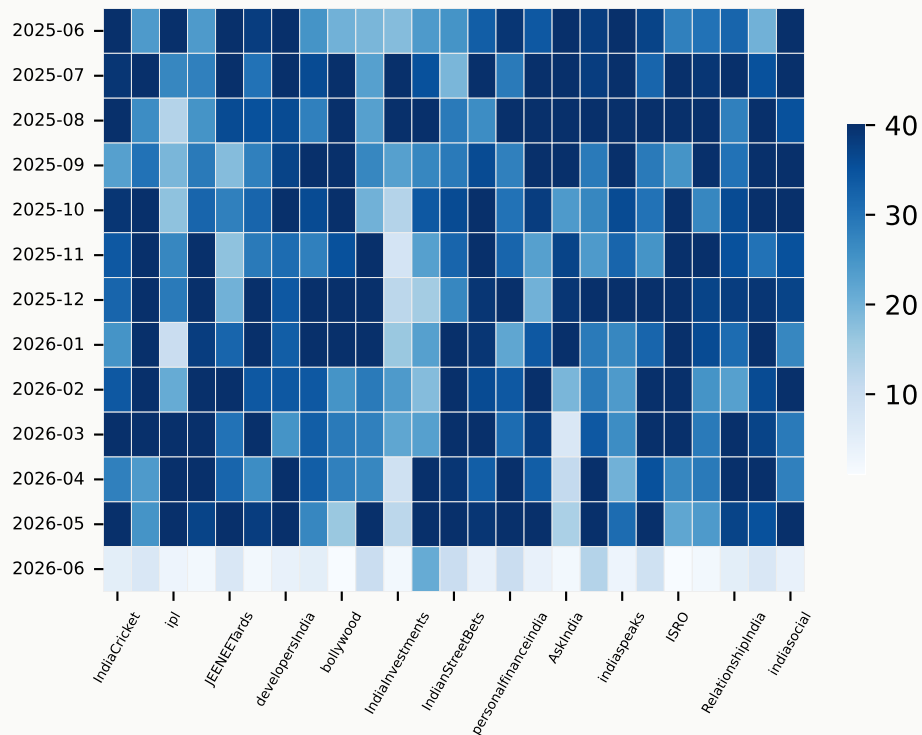
§6 Event Calendar

§7 Findings & Calibrated Weights

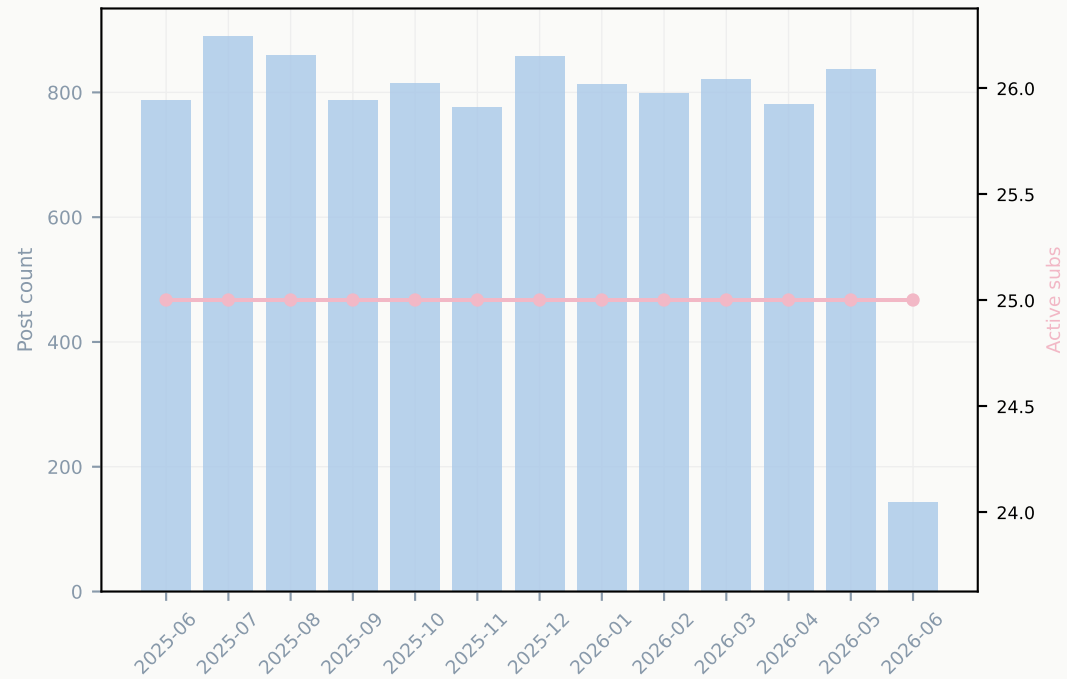
§1 Data Portrait

25 subreddits · 13 months · 9,971 posts · 31,204 comment rows

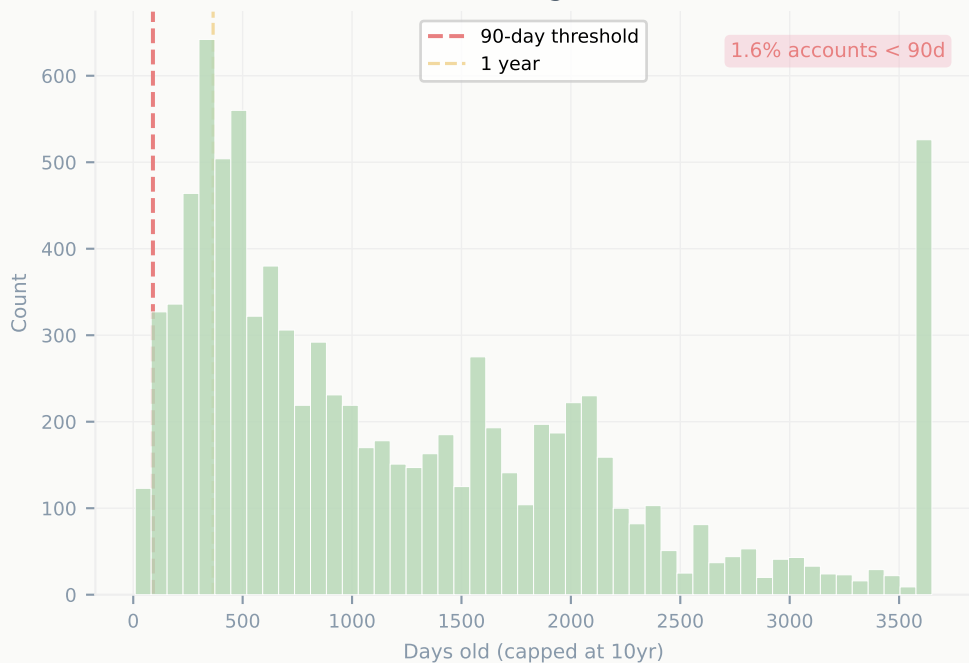
Posts per Sub per Month



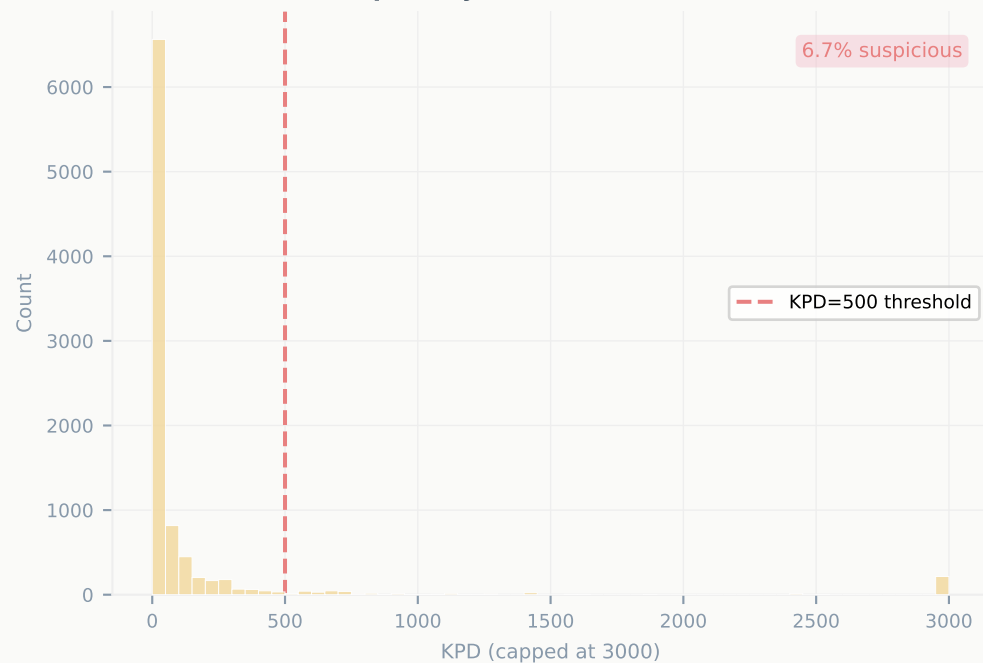
Posts per Month



Poster Account Age Distribution

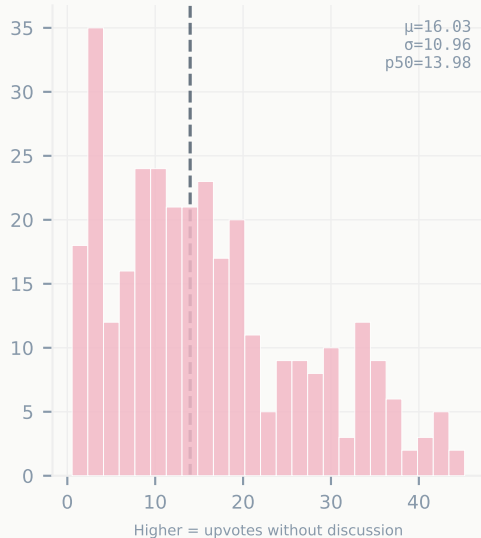
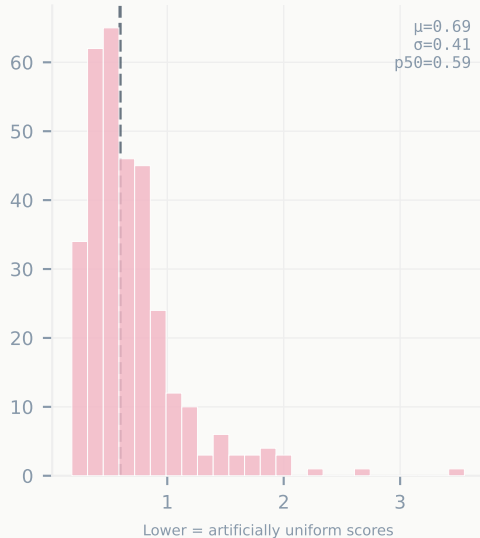
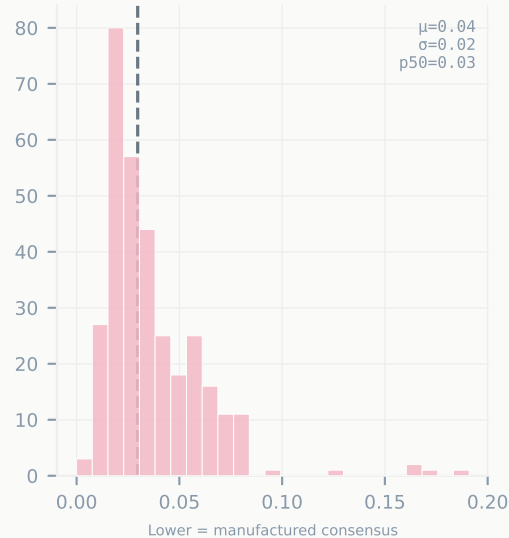
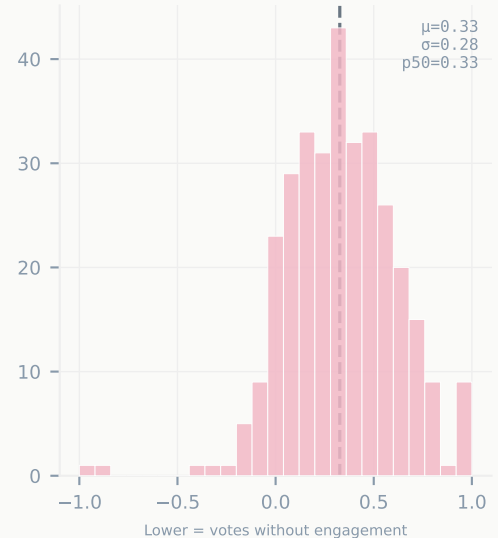
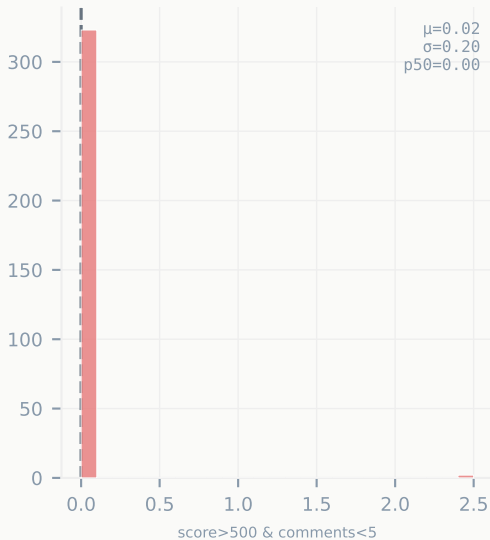
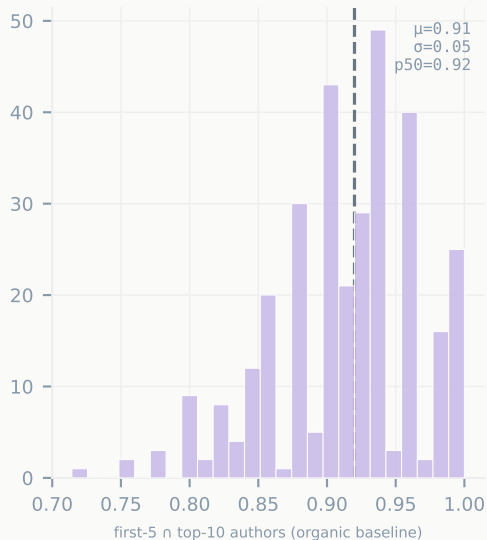
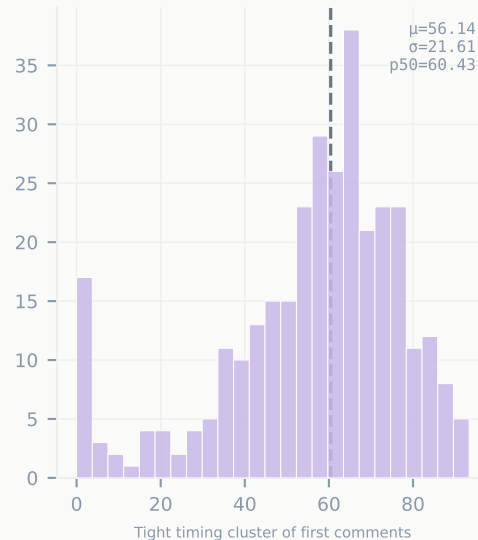
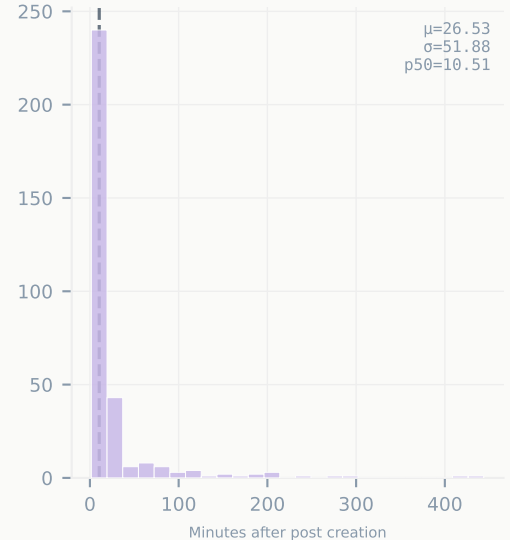


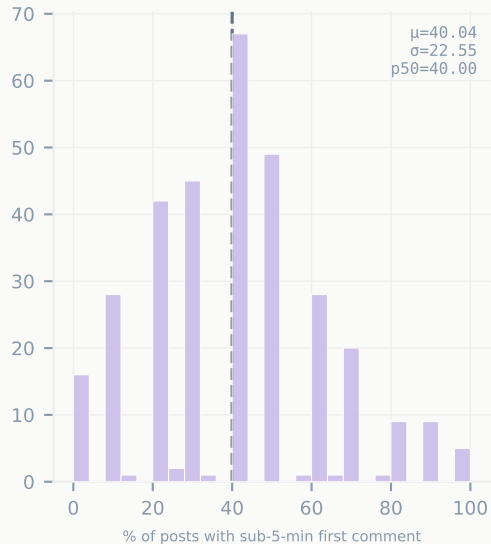
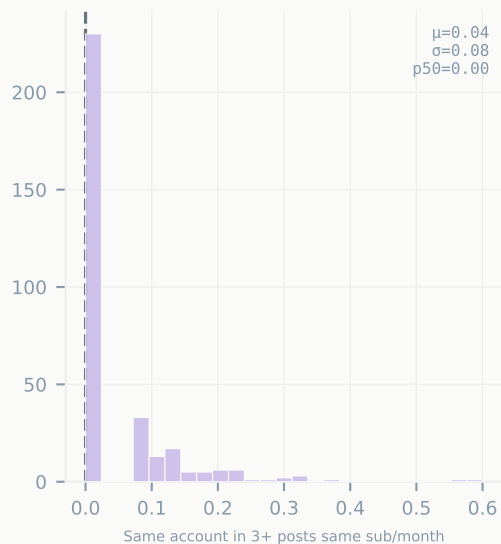
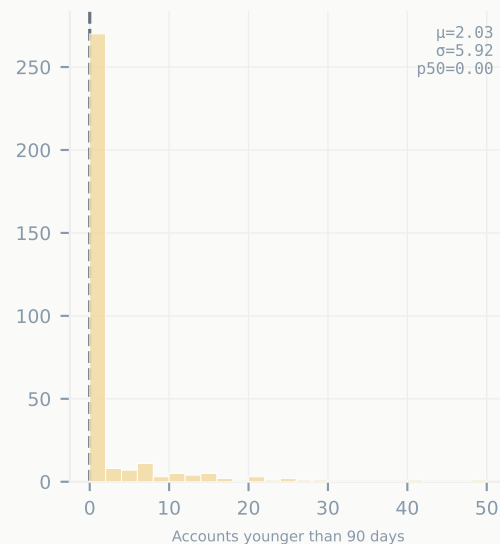
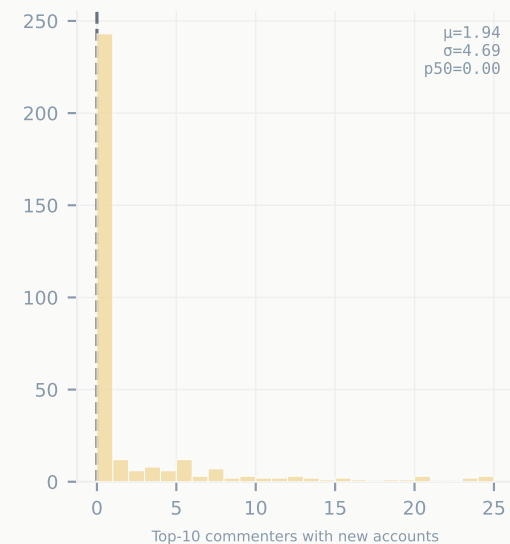
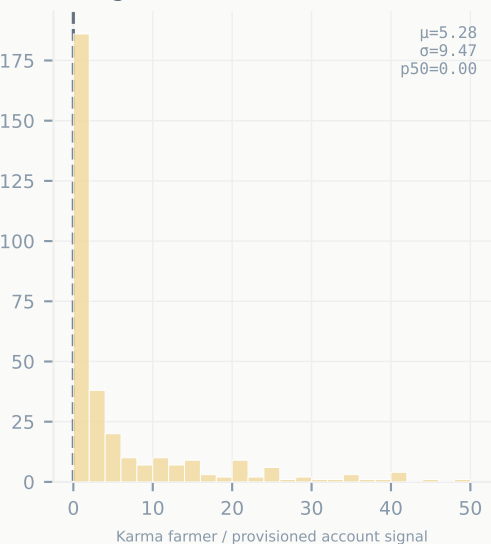
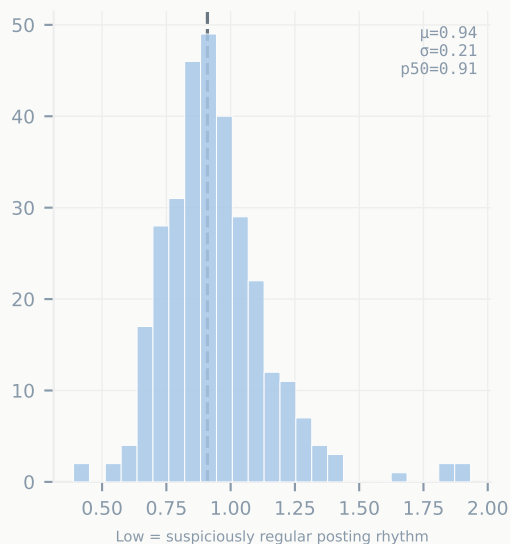
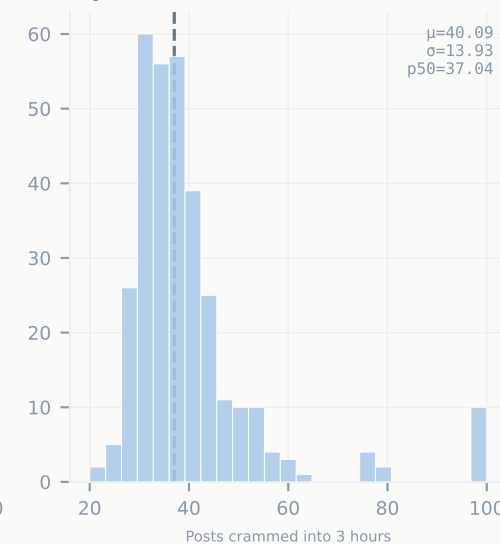
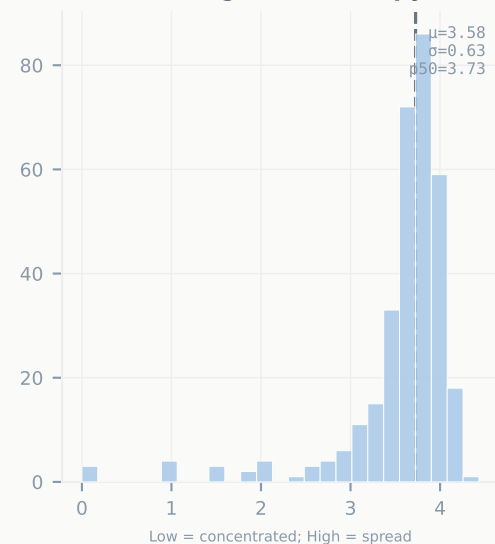
Karma per Day (KPD) — Poster Accounts



§2 Univariate Analysis

Distribution of every raw signal across all sub-month observations

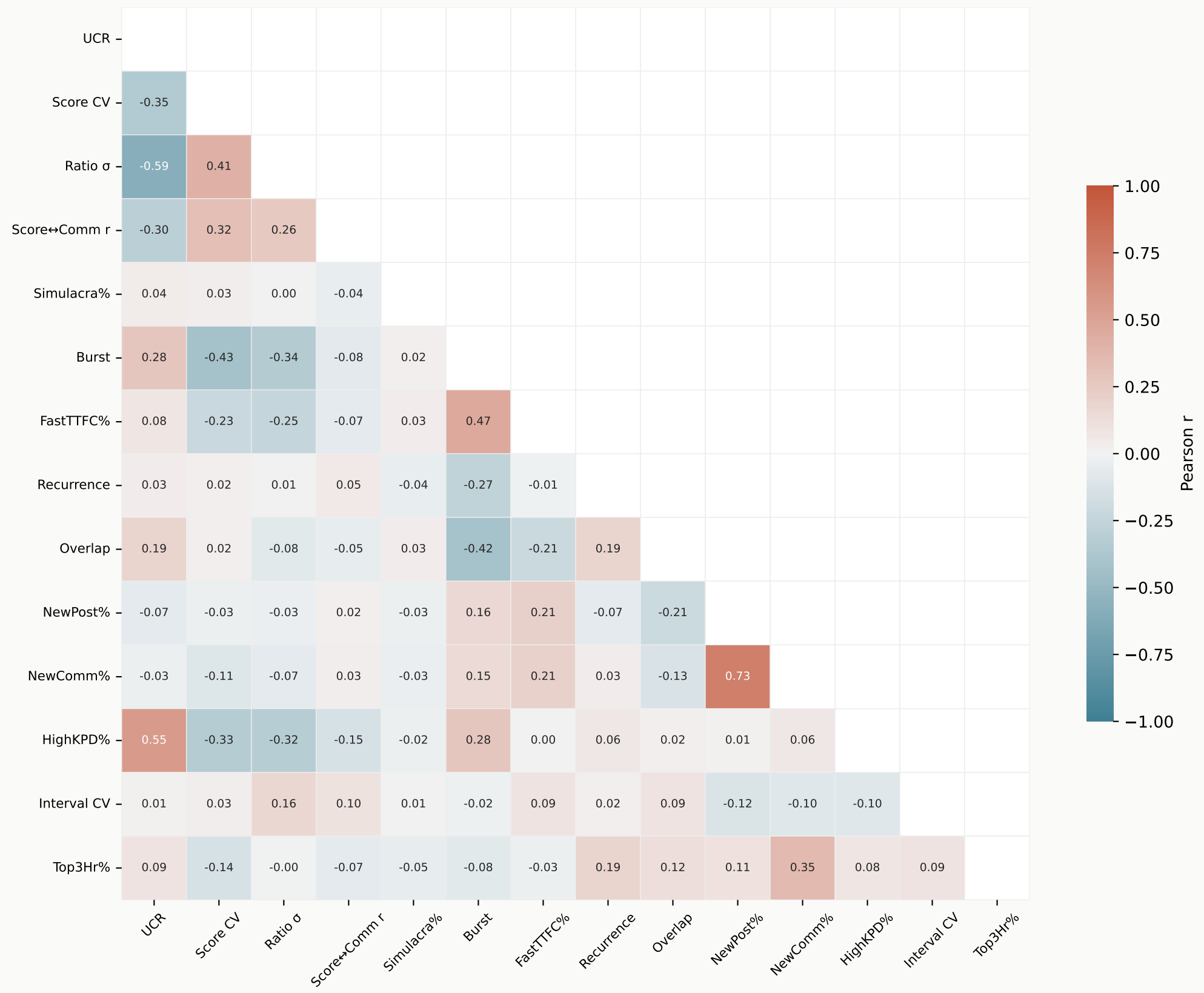
Upvote:Comment Ratio (UCR)**Score Coefficient of Variation****Upvote Ratio Std Dev****Score-Comment Correlation****Simulacra Rate %****Comment Overlap Rate****Burst Score****Avg Time-to-First-Comment (min)**

Fast TTFC % (<5 min)**Commenter Recurrence Rate****New Poster % (<90d)****New Commenter % (<90d)****High-KPD Account % (>500)****Post Interval CV****Top-3 Hour Concentration % (IST)****Posting Hour Entropy**

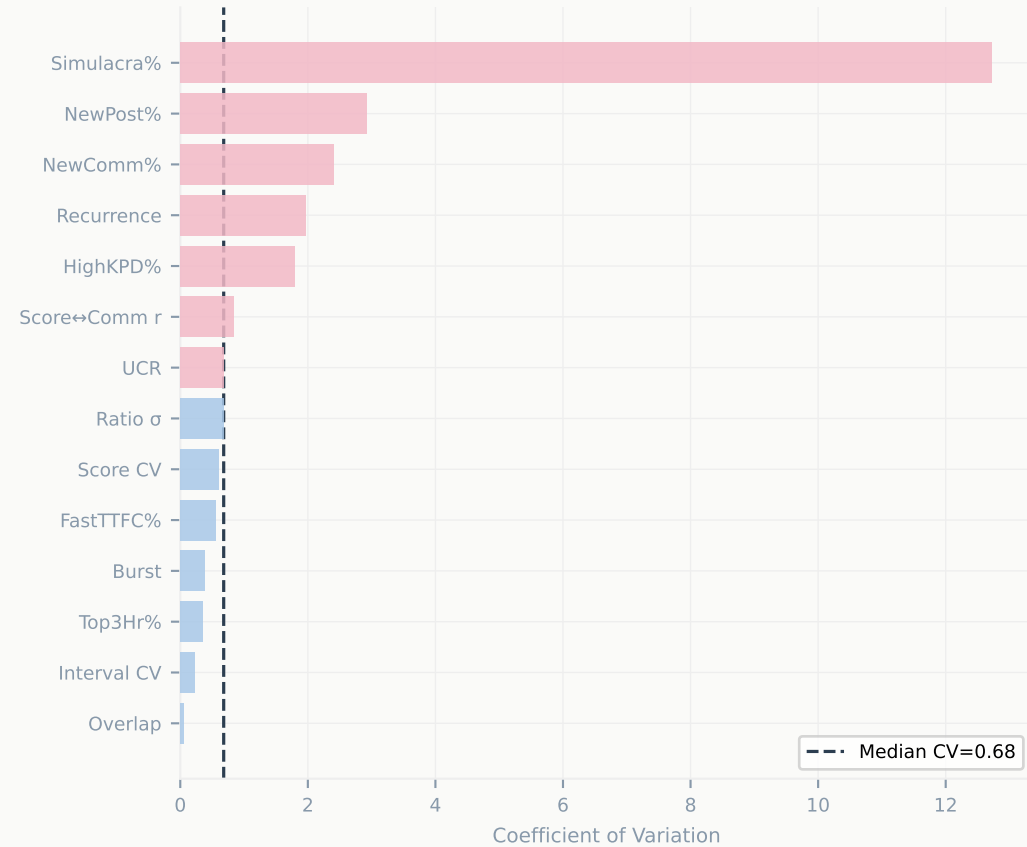
§3 Signal Correlations

Collinearity between signals — and which signals discriminate best across subreddits

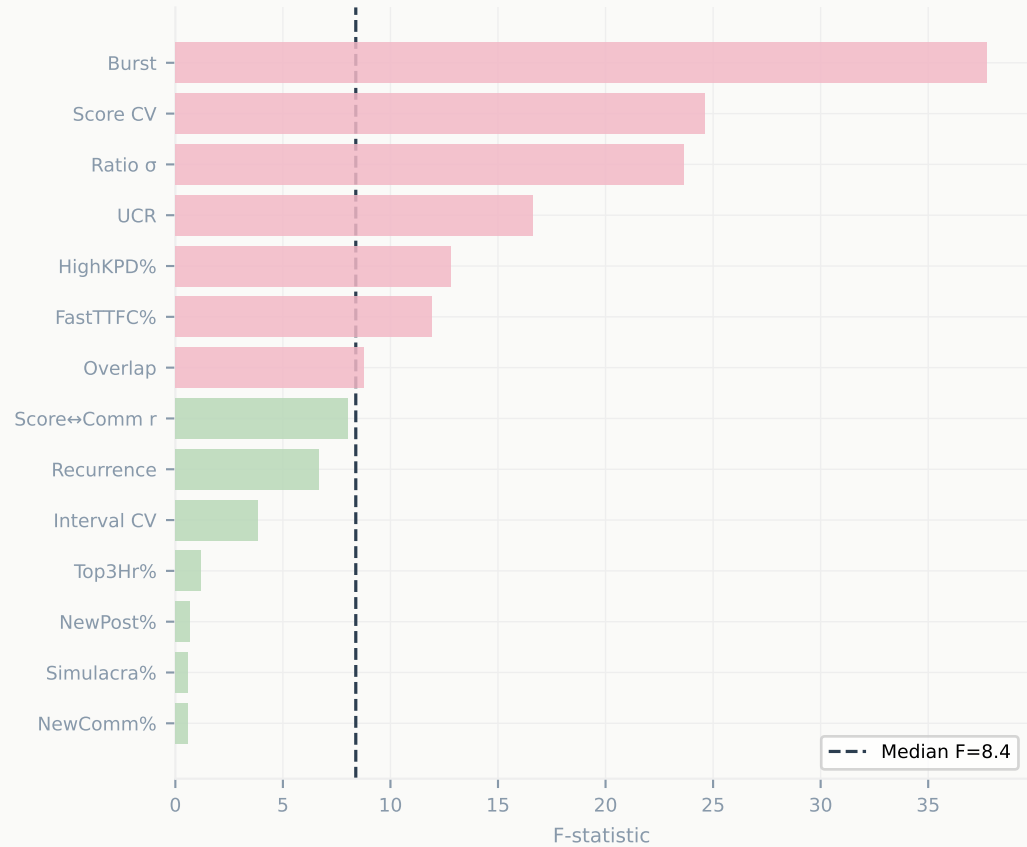
Signal Pairwise Correlation Matrix (lower triangle)



Discriminative Power (CV across sub-months)
Higher = better at differentiating subs



Between-Cluster Discrimination (ANOVA F)
Higher = signal differs meaningfully across sub types

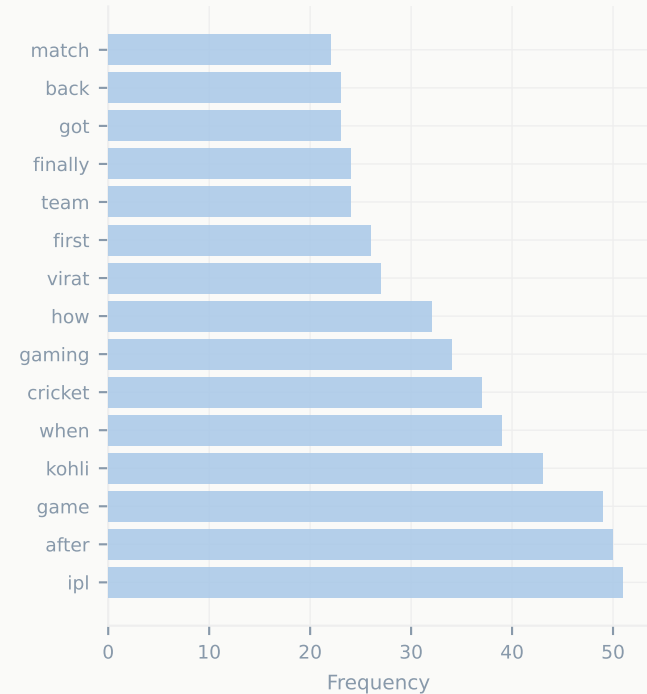


§4 Text Intelligence

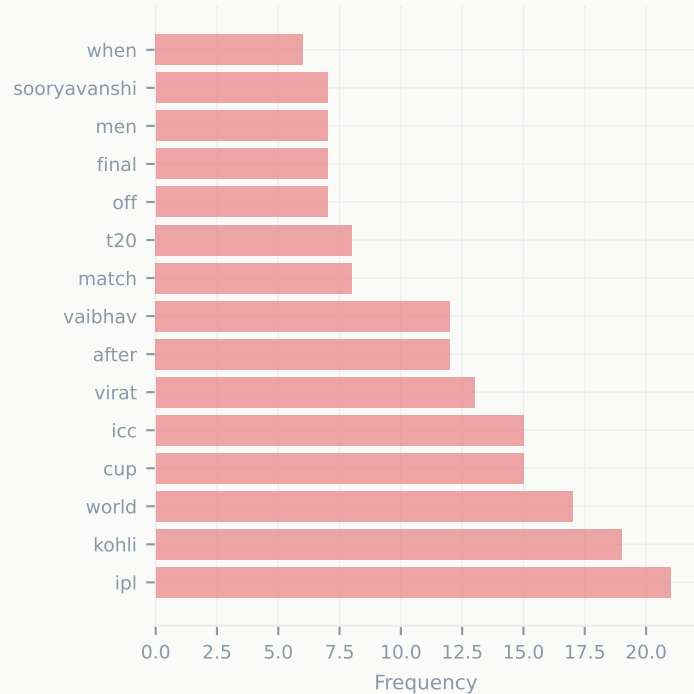
Topics, narrative pushes, near-duplicates, cross-sub keyword infiltration — from post titles

Cluster: Cricket/Sports (1,161 posts across 3 subs)

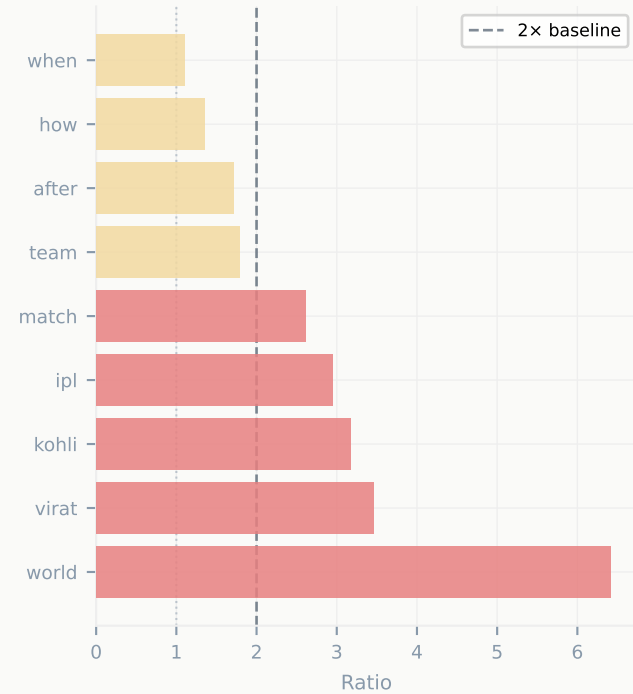
Top Words — Organic Accounts



Top Words — Suspicious Accounts (142 posts)

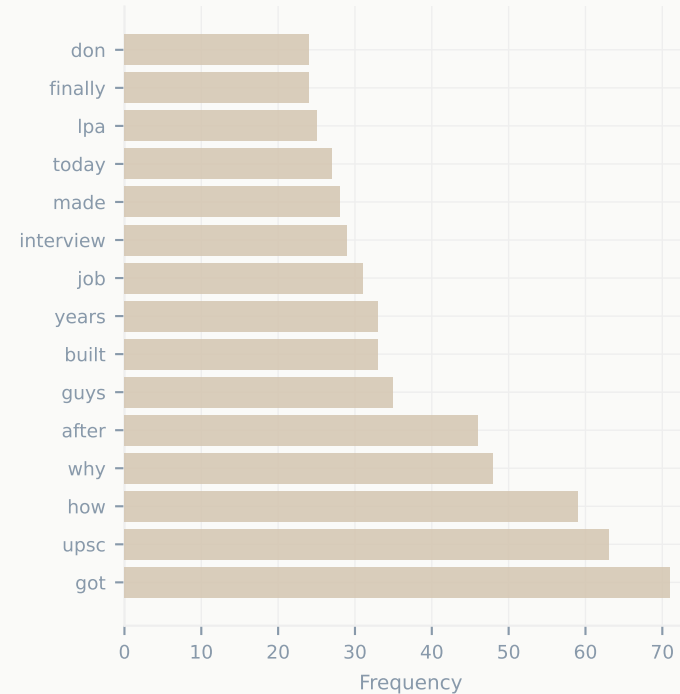


Pushed Narratives (suspicious/organic freq ratio)

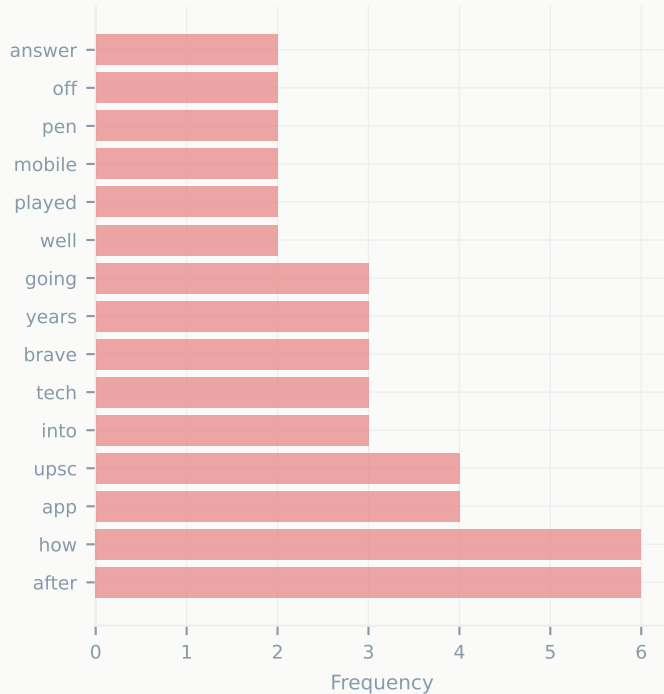


Cluster: Education (1,641 posts across 4 subs)

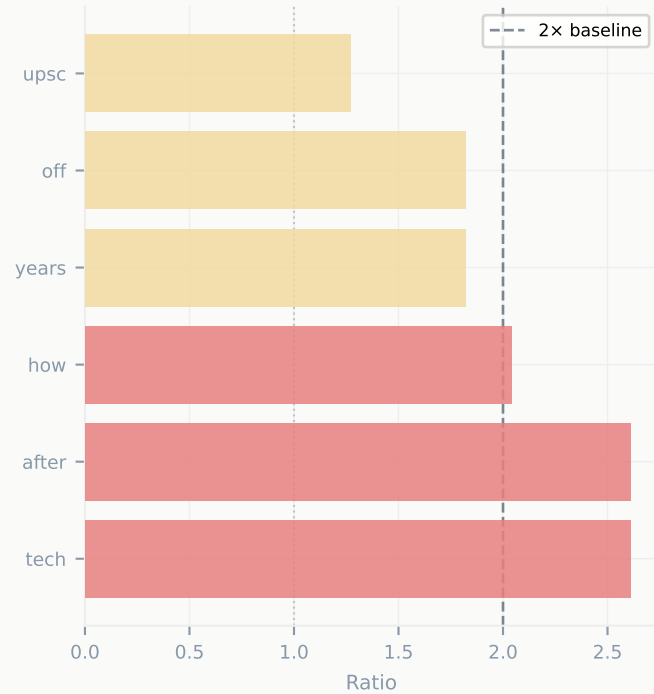
Top Words — Organic Accounts



Top Words — Suspicious Accounts (78 posts)

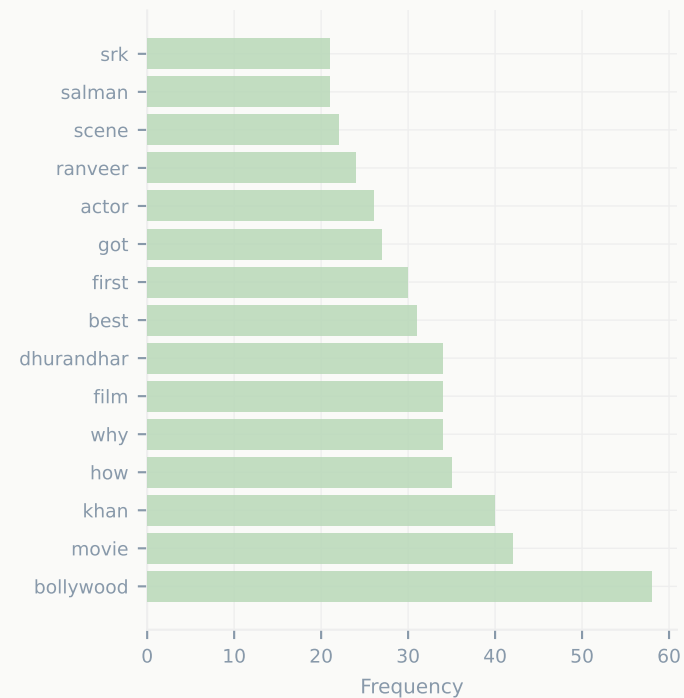


Pushed Narratives (suspicious/organic freq ratio)

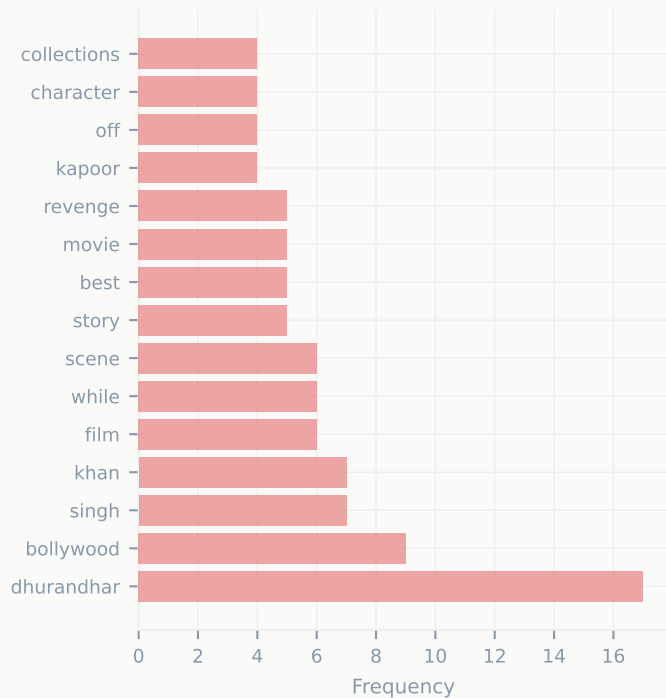


Cluster: Entertainment (1,165 posts across 3 subs)

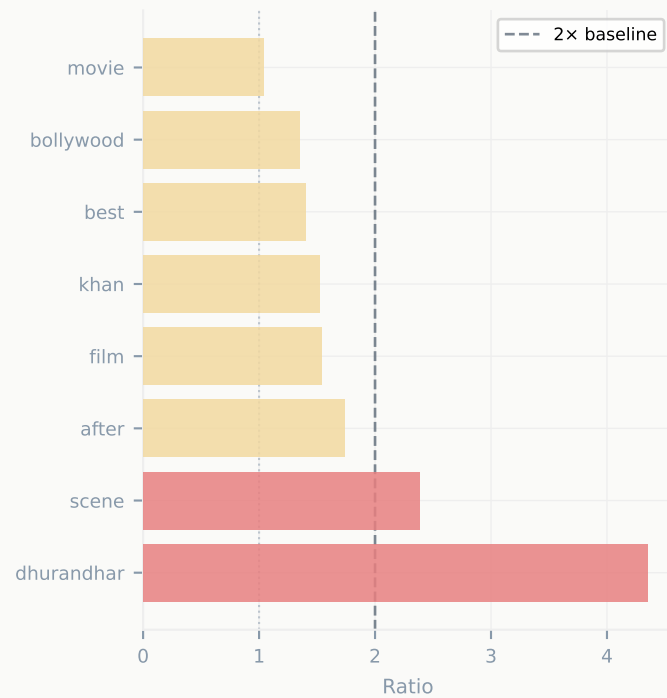
Top Words — Organic Accounts



Top Words — Suspicious Accounts (120 posts)

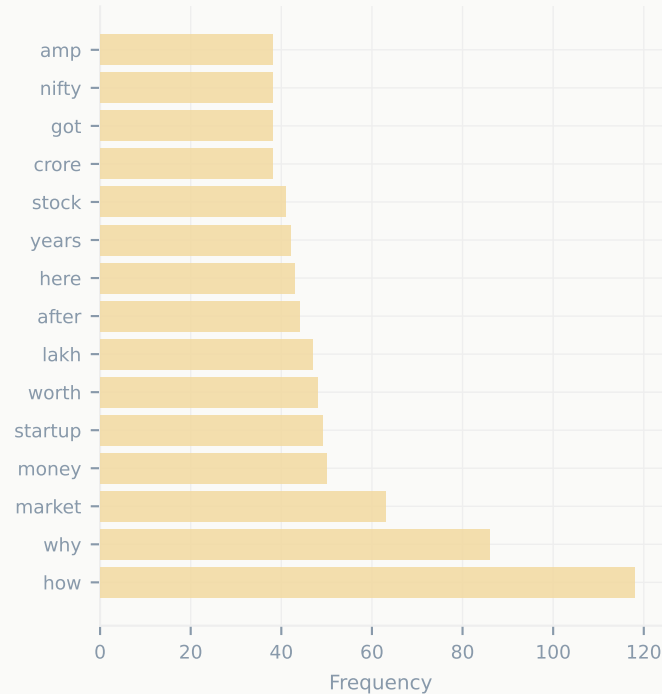


Pushed Narratives (suspicious/organic freq ratio)

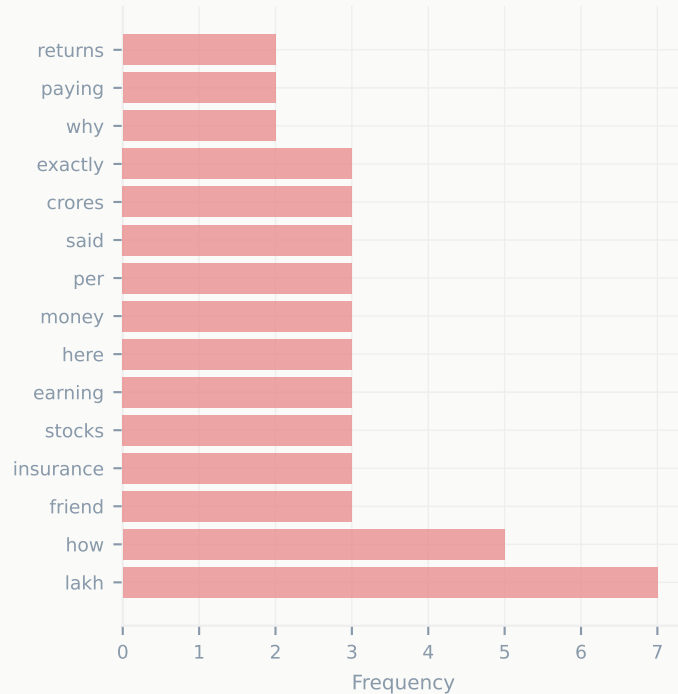


Cluster: Finance (1,868 posts across 5 subs)

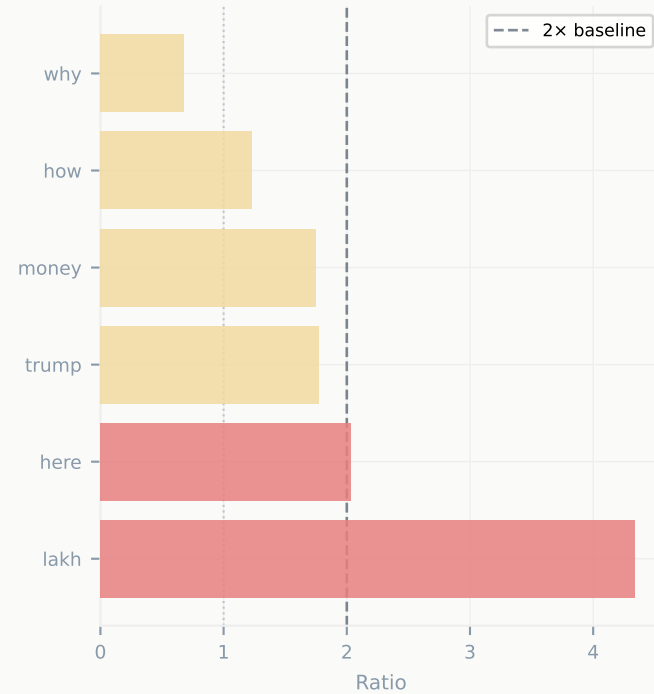
Top Words — Organic Accounts



Top Words — Suspicious Accounts (62 posts)

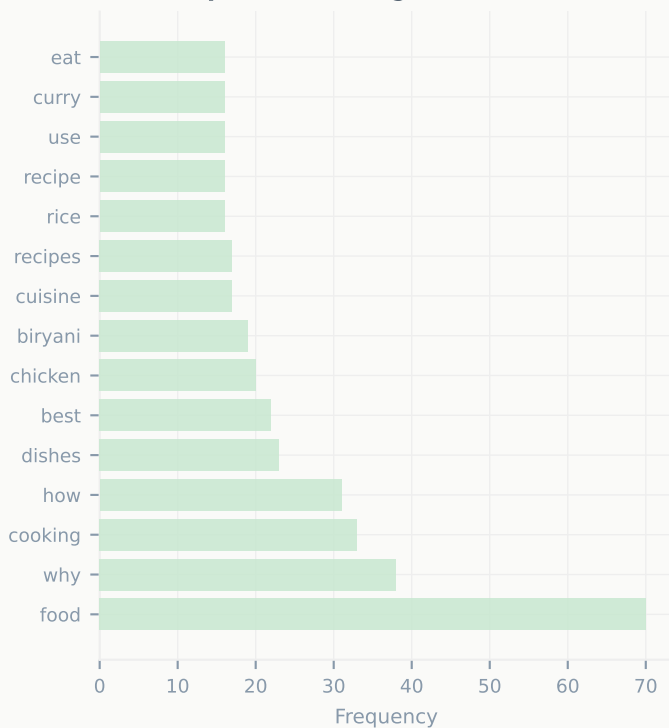


Pushed Narratives (suspicious/organic freq ratio)

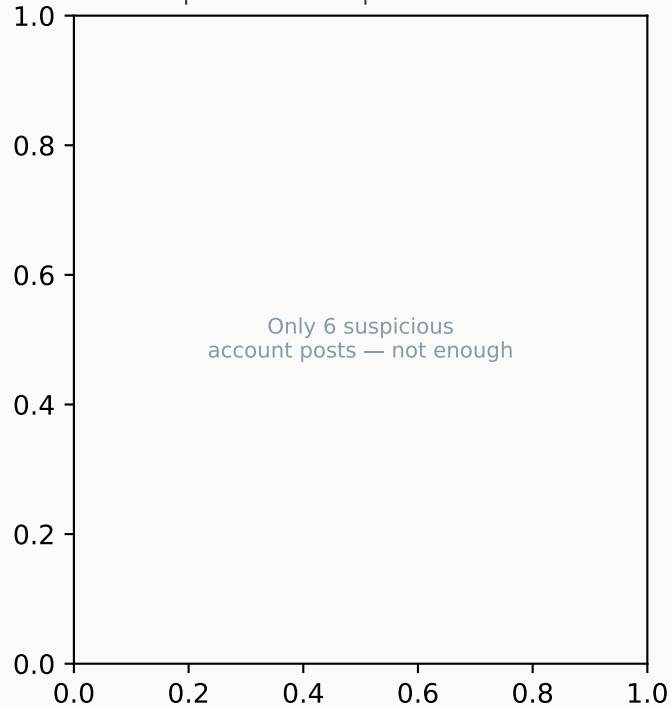


Cluster: Lifestyle (424 posts across 1 subs)

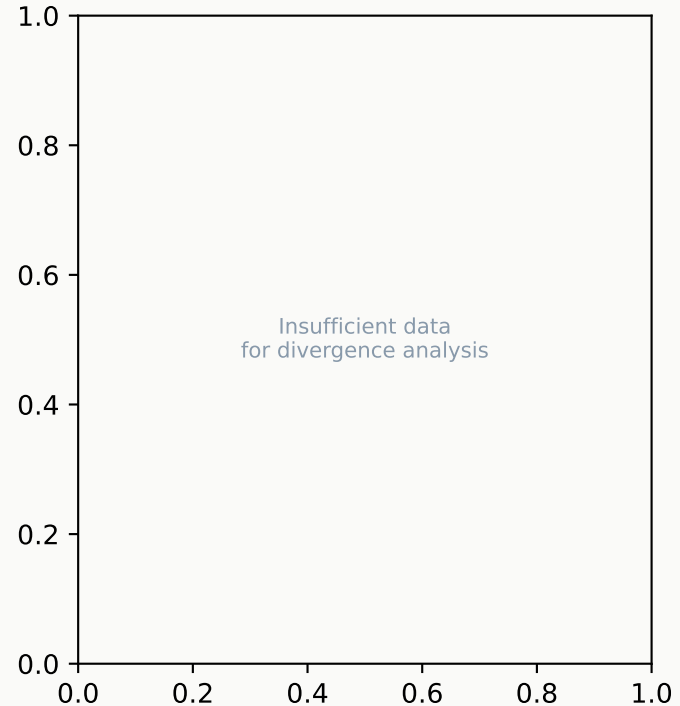
Top Words — Organic Accounts



Top Words — Suspicious Accounts

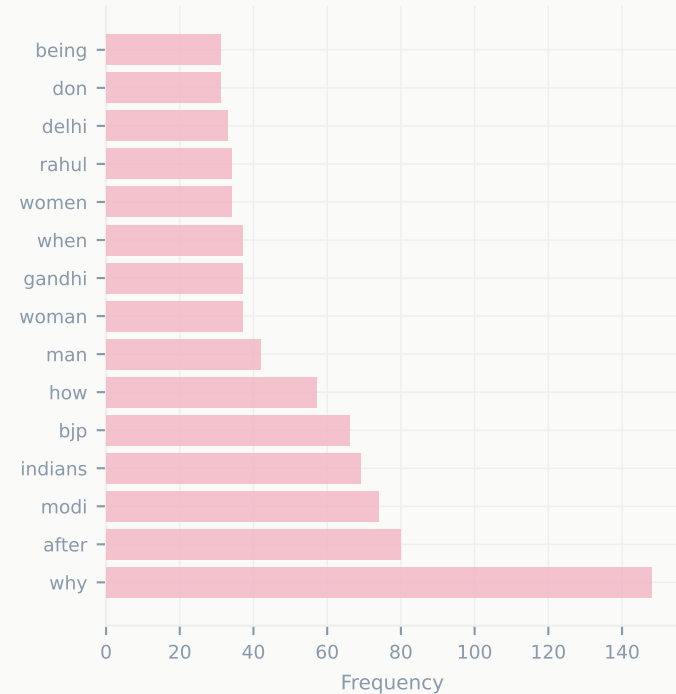


Pushed Narratives

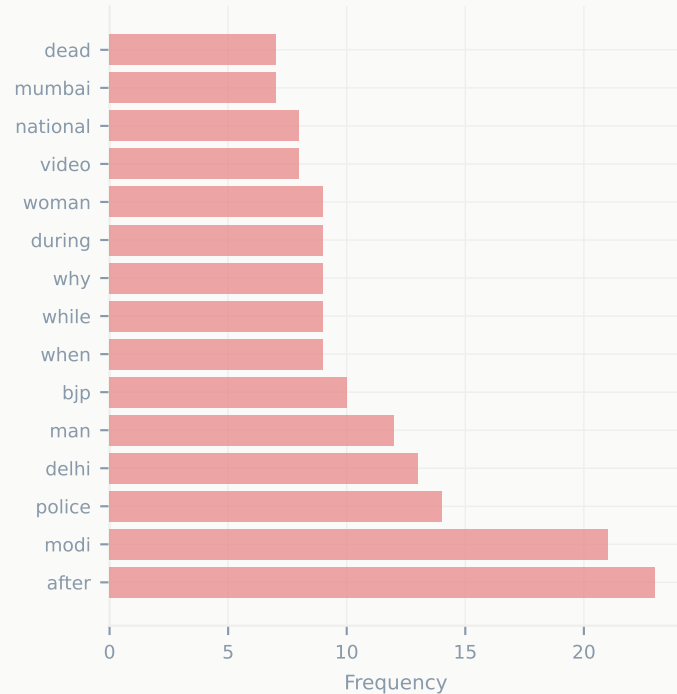


Cluster: Political (1,602 posts across 4 subs)

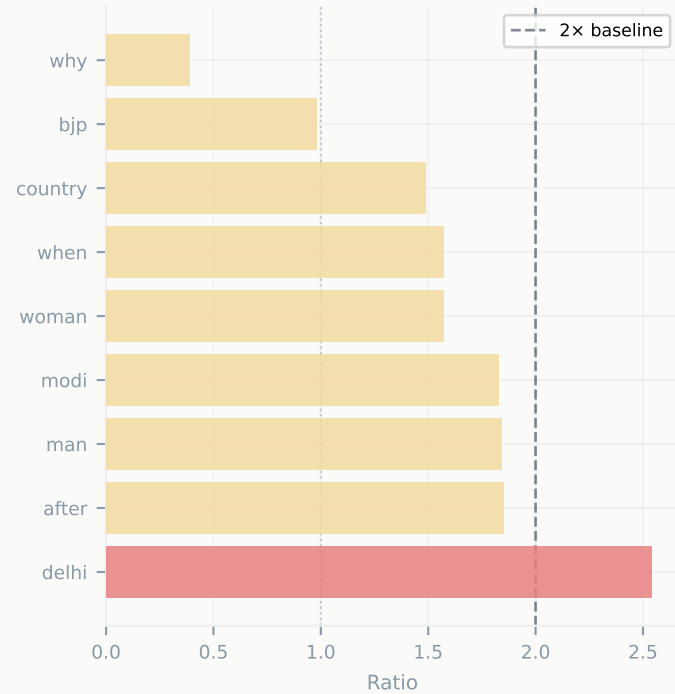
Top Words — Organic Accounts



Top Words — Suspicious Accounts (215 posts)

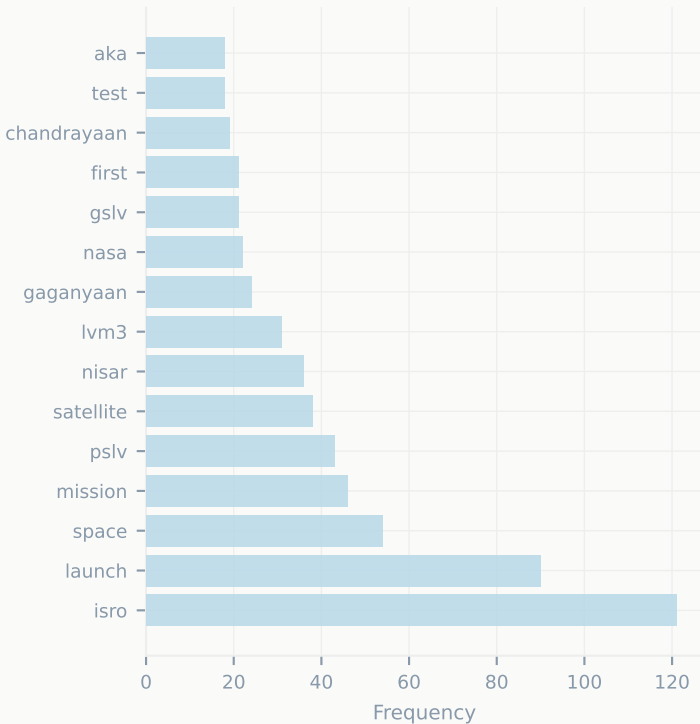


Pushed Narratives (suspicious/organic freq ratio)

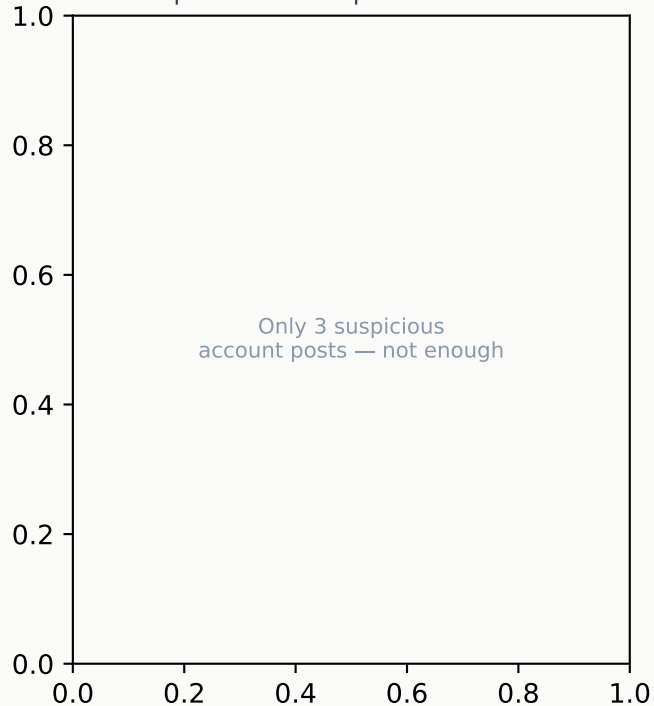


Cluster: Science (423 posts across 1 subs)

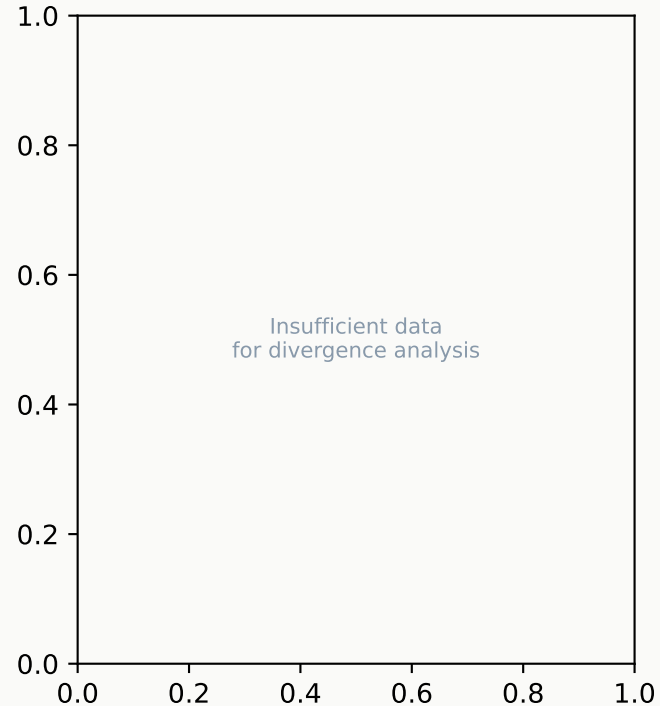
Top Words — Organic Accounts



Top Words — Suspicious Accounts

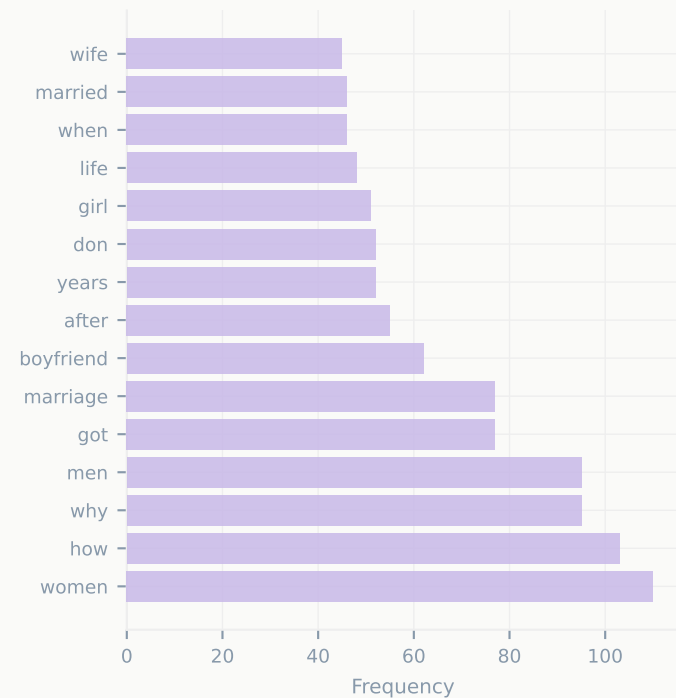


Pushed Narratives

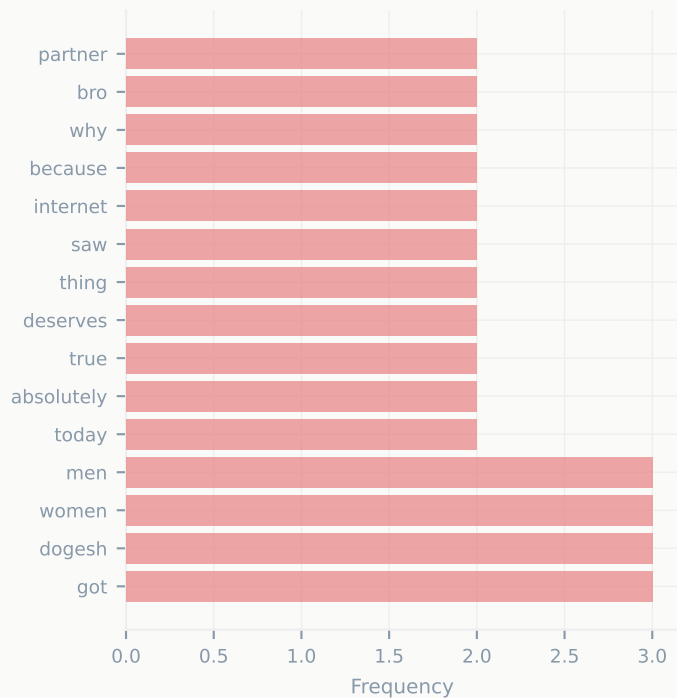


Cluster: Women/Social (1,687 posts across 4 subs)

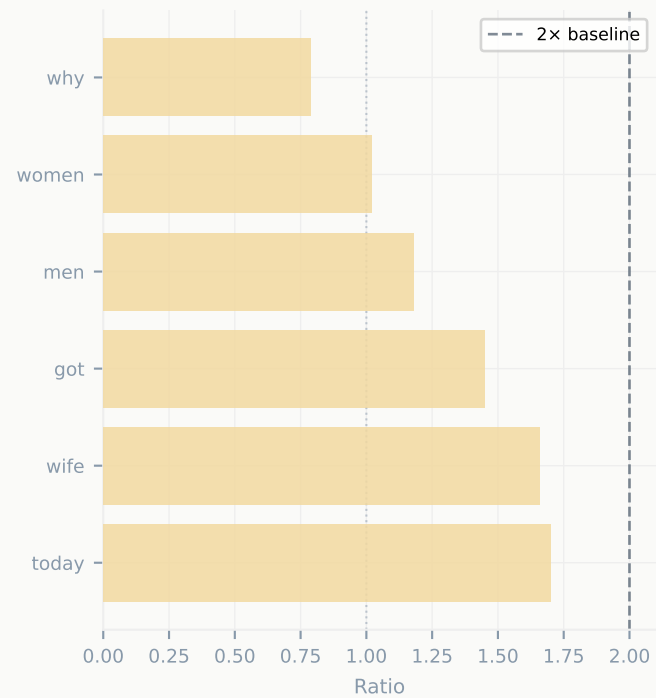
Top Words — Organic Accounts



Top Words — Suspicious Accounts (44 posts)

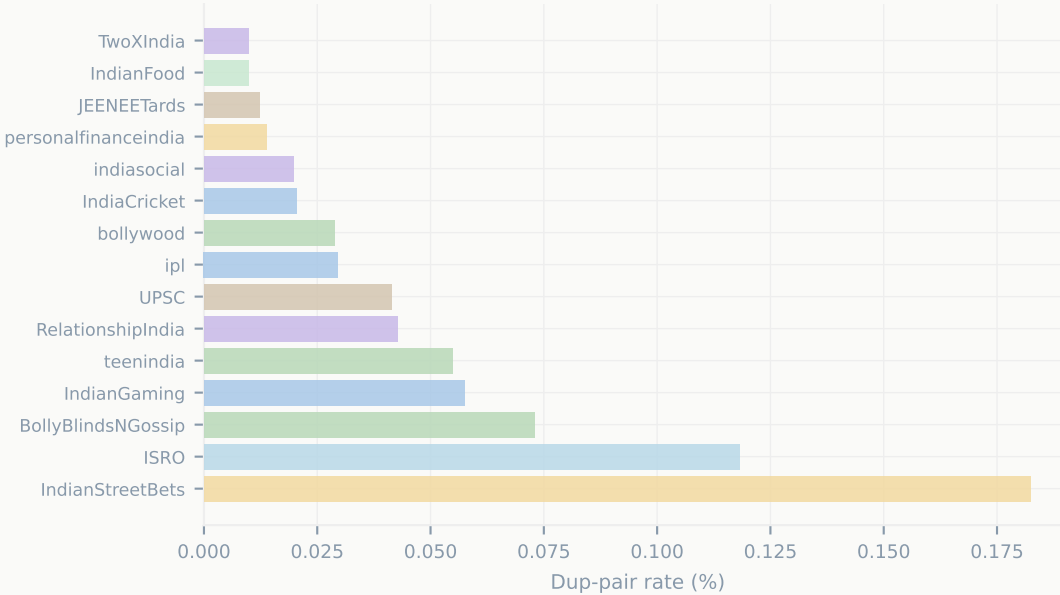


Pushed Narratives (suspicious/organic freq ratio)

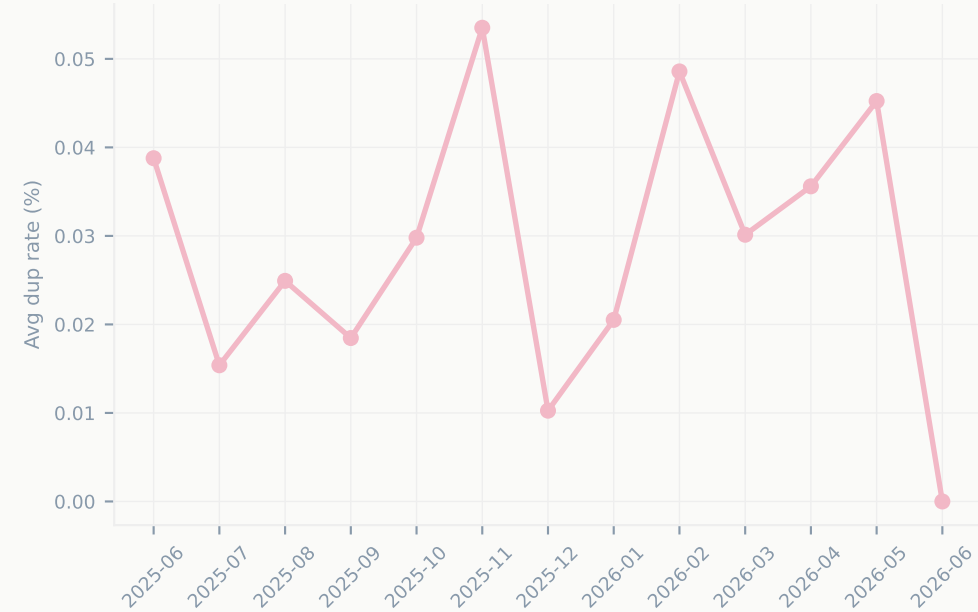


Near-Duplicate Post Detection (Jaccard similarity on title word sets > 0.5)

Avg Near-Duplicate Rate by Sub



Near-Duplicate Rate Over Time



Example Near-Duplicate Pairs

r/BollyBlindsNGossip [2026-03] sim=0.62
A: Gaurav Kapur and Kritika Kamra got married at their bandra home.
B: Gaurav Kapur and Kritika Kamra are now married.

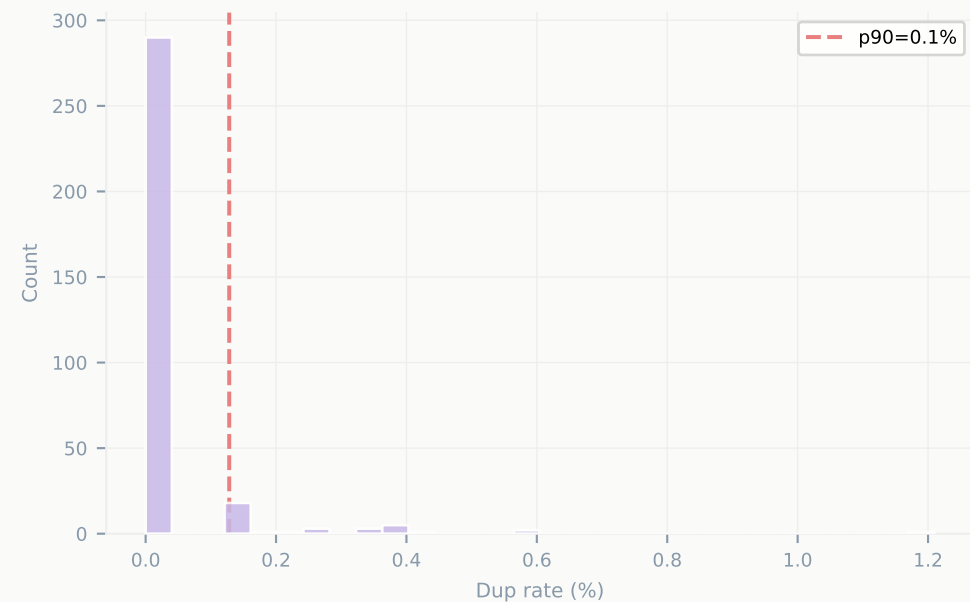
r/BollyBlindsNGossip [2026-03] sim=0.67
A: Kritika Kamra and Gaurav Kapur at their Reception.
B: Gaurav Kapur and Kritika Kamra are now married.

r/BollyBlindsNGossip [2026-05] sim=0.75
A: Alia Bhatt at Cannes 2026.
B: Alia Bhatt's first look of Cannes.

r/BollyBlindsNGossip [2026-05] sim=0.75
A: Aishwarya rai second day at the CANNES film festival red carpet
B: Aishwarya rai's second look at Cannes film festival

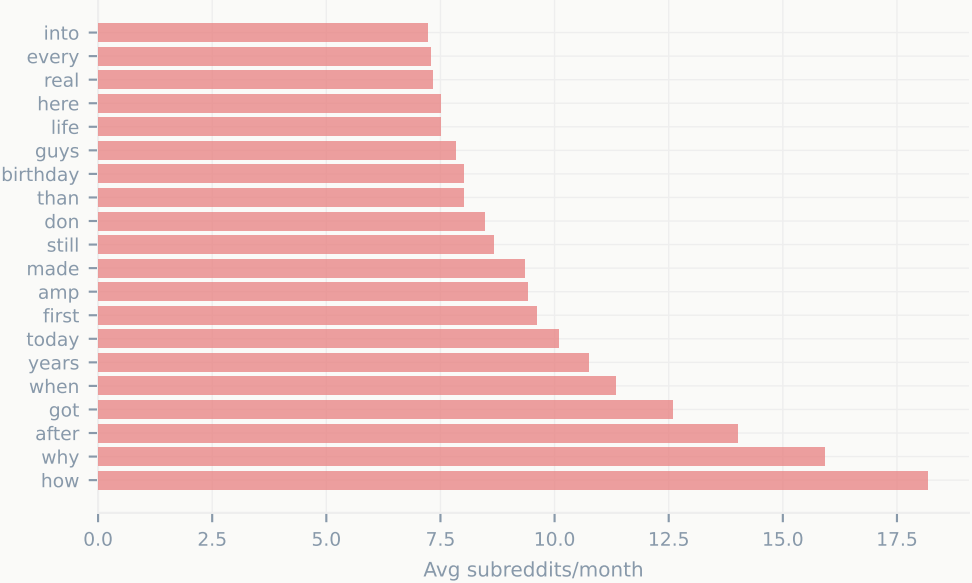
r/ISRO [2025-07] sim=0.8
A: GSVL-F16 NISAR Liftoff
B: GSVL-F16/NISAR liftoff images

Distribution of Near-Dup Rates All sub-months

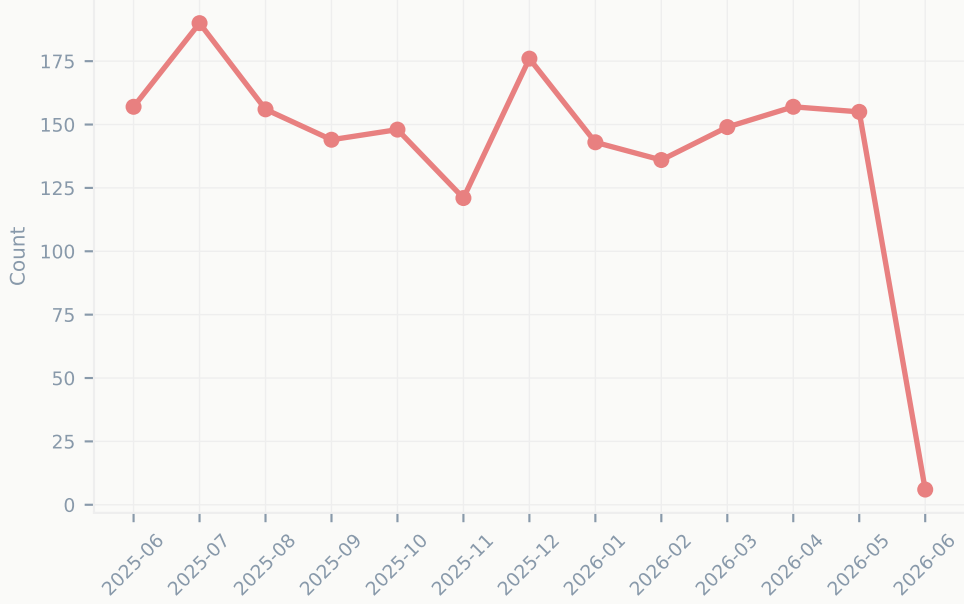


Cross-Sub Keyword Infiltration (words appearing in 4+ subreddits in same month)

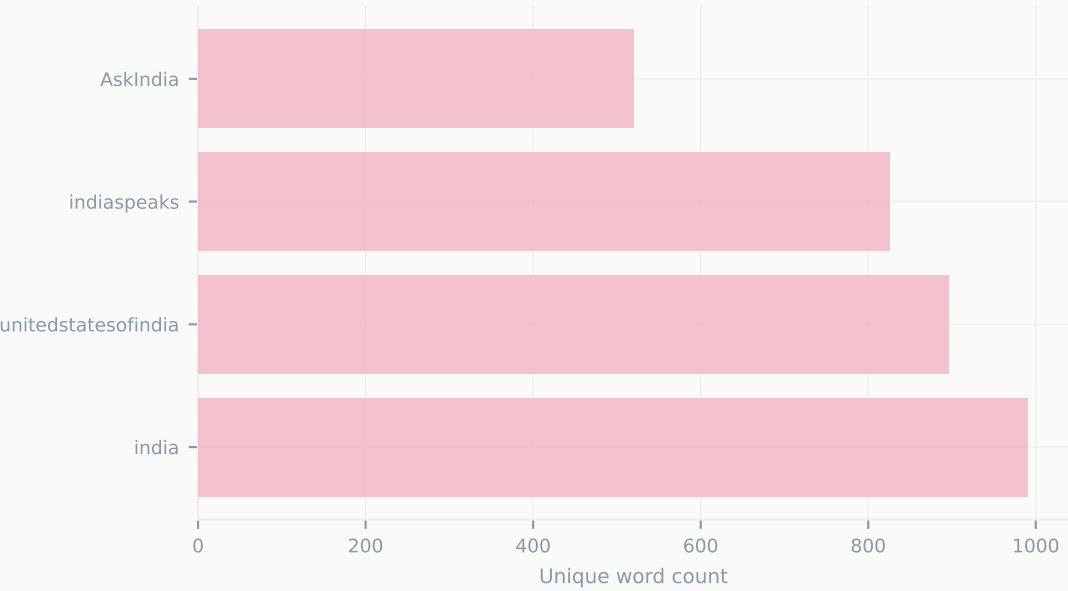
Most Cross-Sub Keywords (avg subs/month)



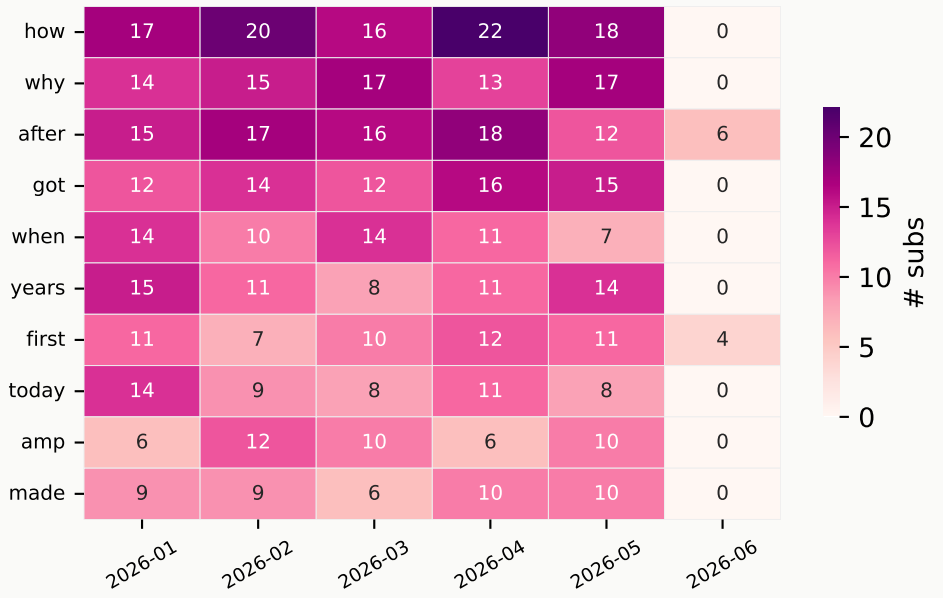
Cross-Sub Keywords Over Time (# words in 4+ subs per month)



Unique Vocabulary — Political Cluster (words used in this sub not in others)



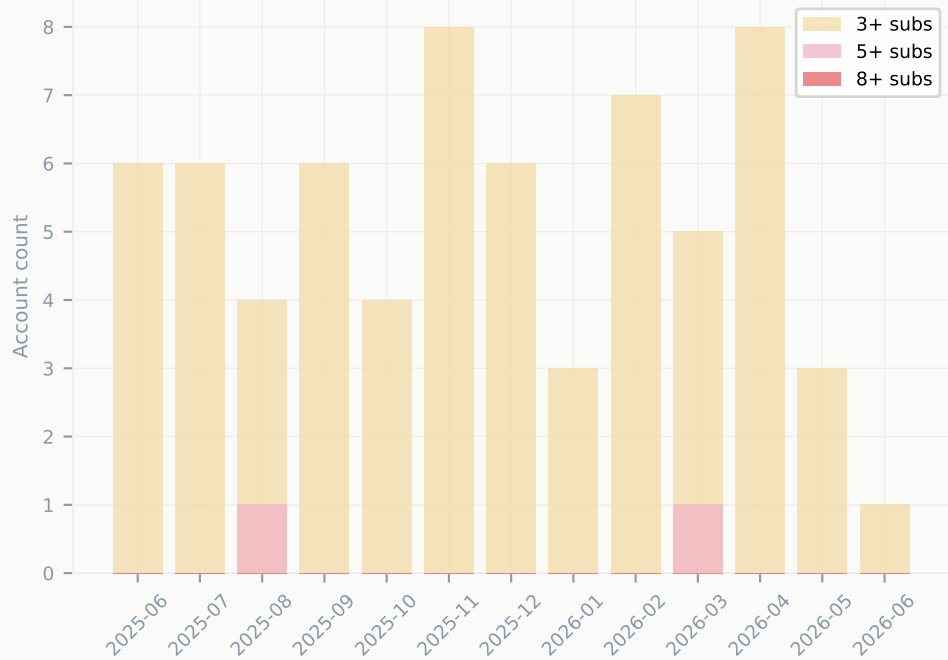
Cross-Sub Word Spread (last 6 months)



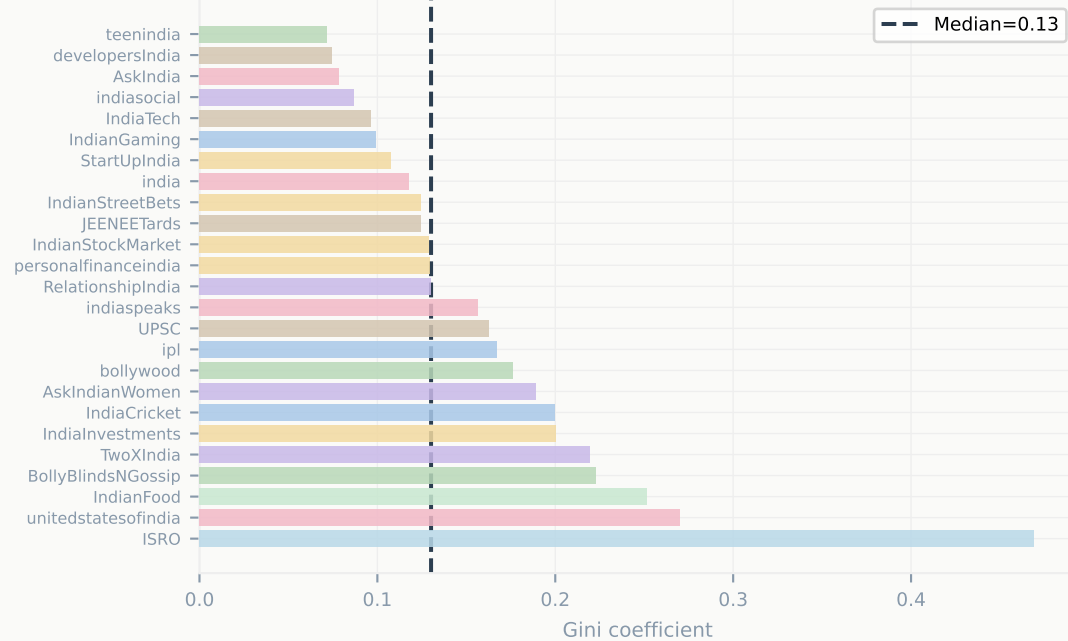
§5 Cross-Sub Network

Same accounts appearing across subreddits — the true ring signal

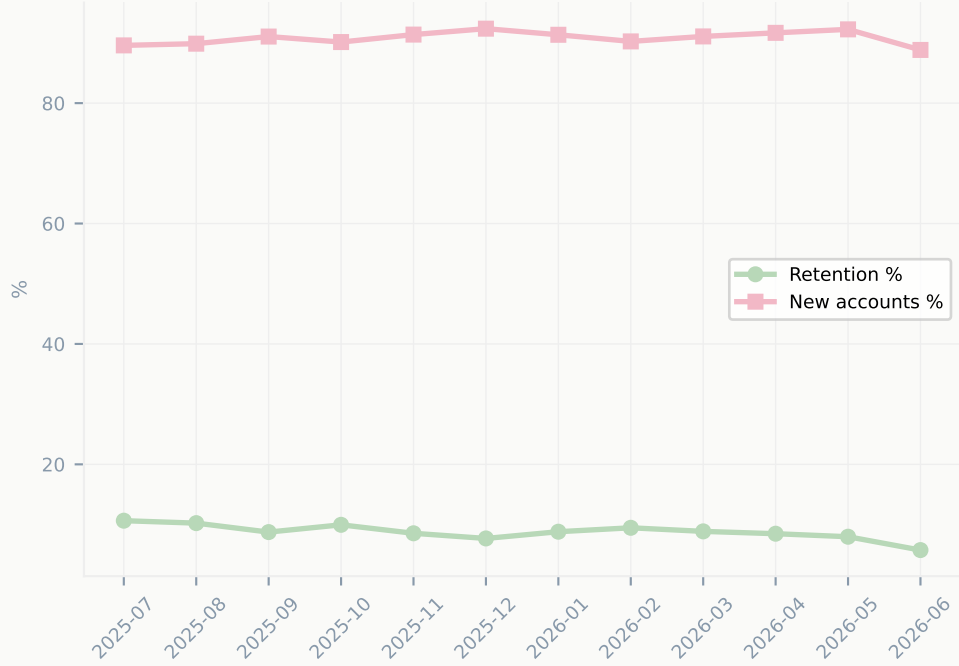
Cross-Sub Commenter Accounts per Month
(accounts appearing in N+ subreddits)



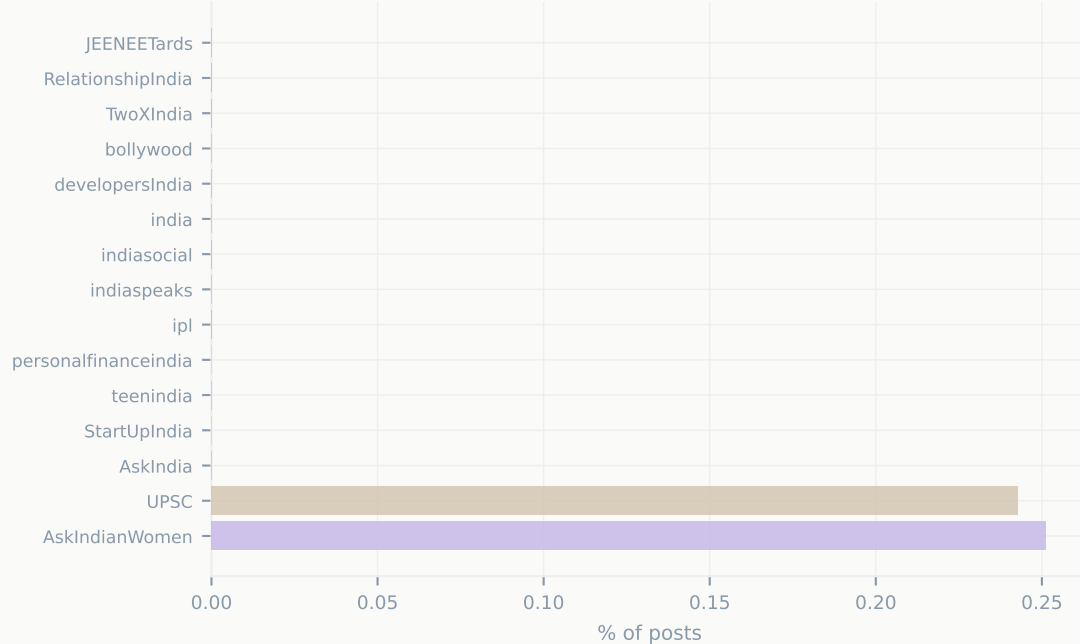
Participation Gini by Sub
(higher = more concentrated commenter activity)



Commenter Churn (month-over-month, all subs)
>60% new each month = possible account cycling

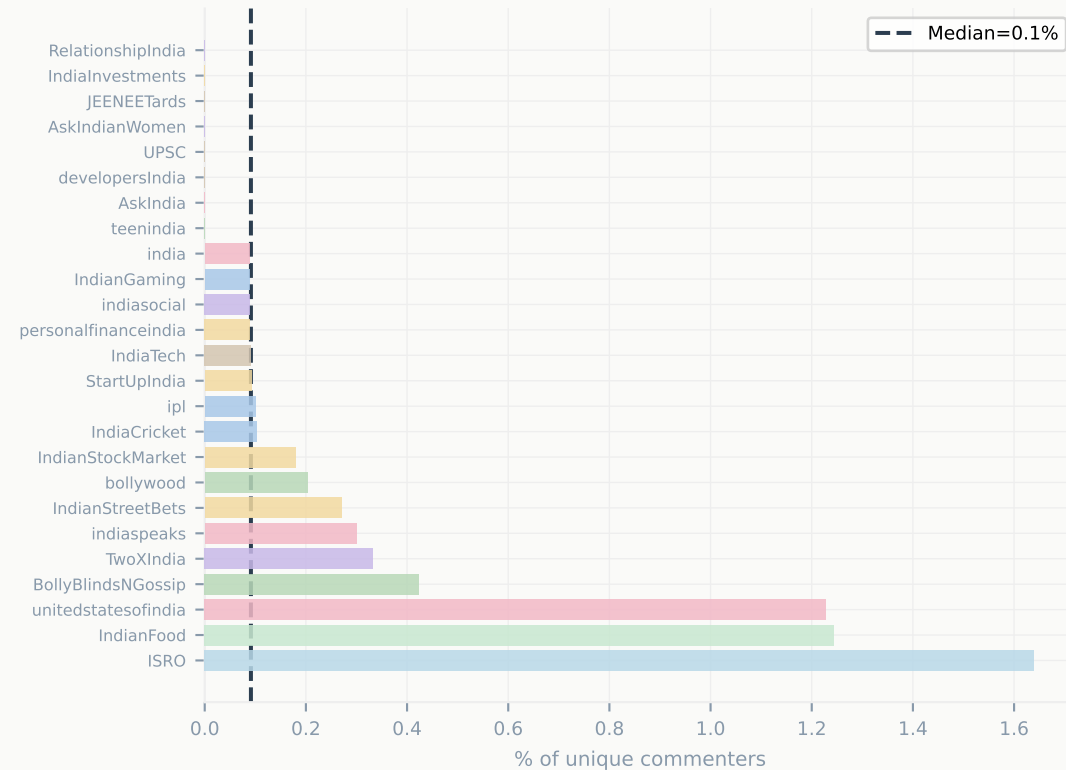


Simulacra Rate by Sub
(score>500 & comments<5 — upvotes with no discussion)

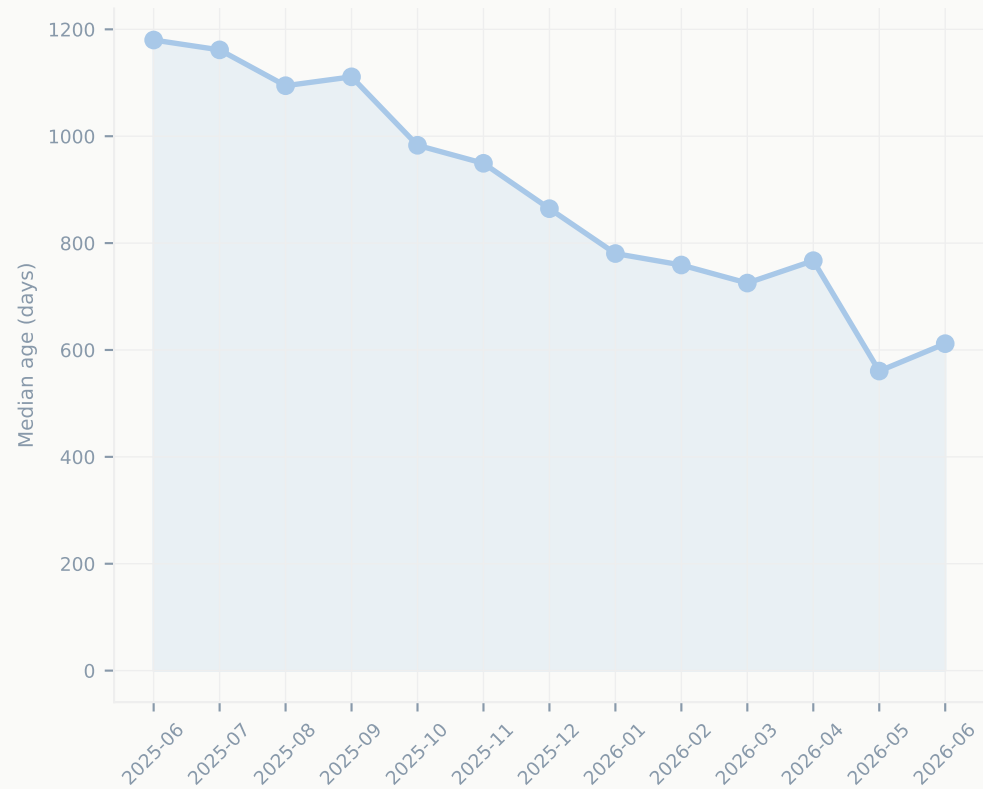


Account Persistence Across 12 Months

Top-Commenter Persistence (present in 6+ months)
High % = genuine community OR entrenched bot ring



Median Poster Account Age Over Time
Sudden drops suggest new-account bot waves

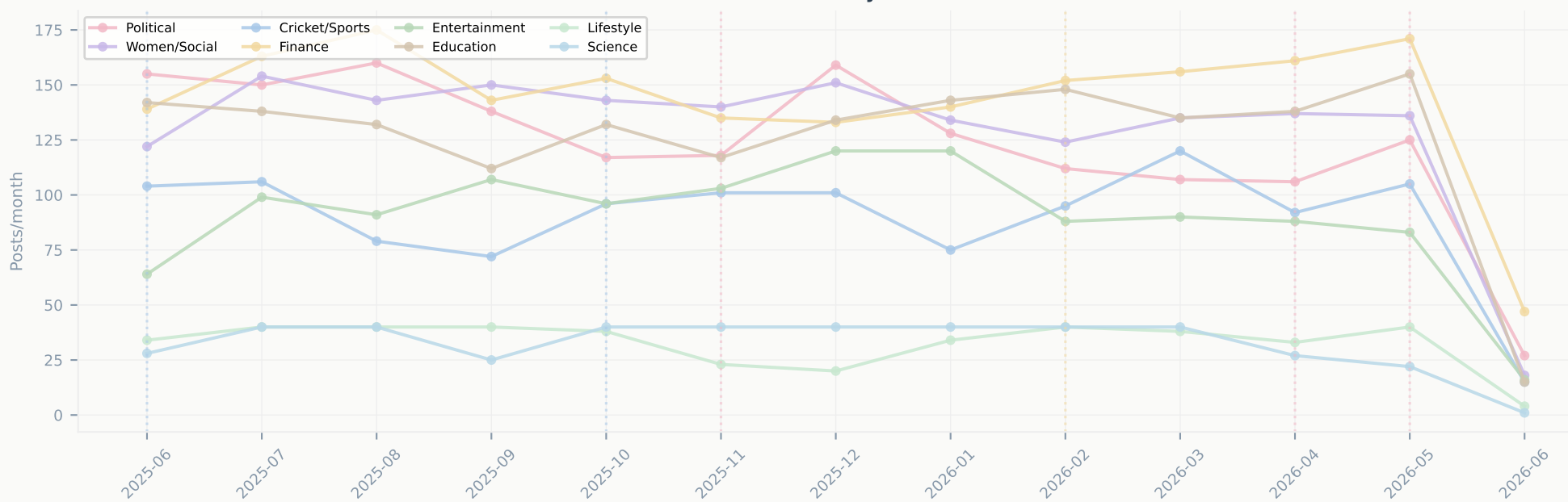


§6 Event Calendar Overlay

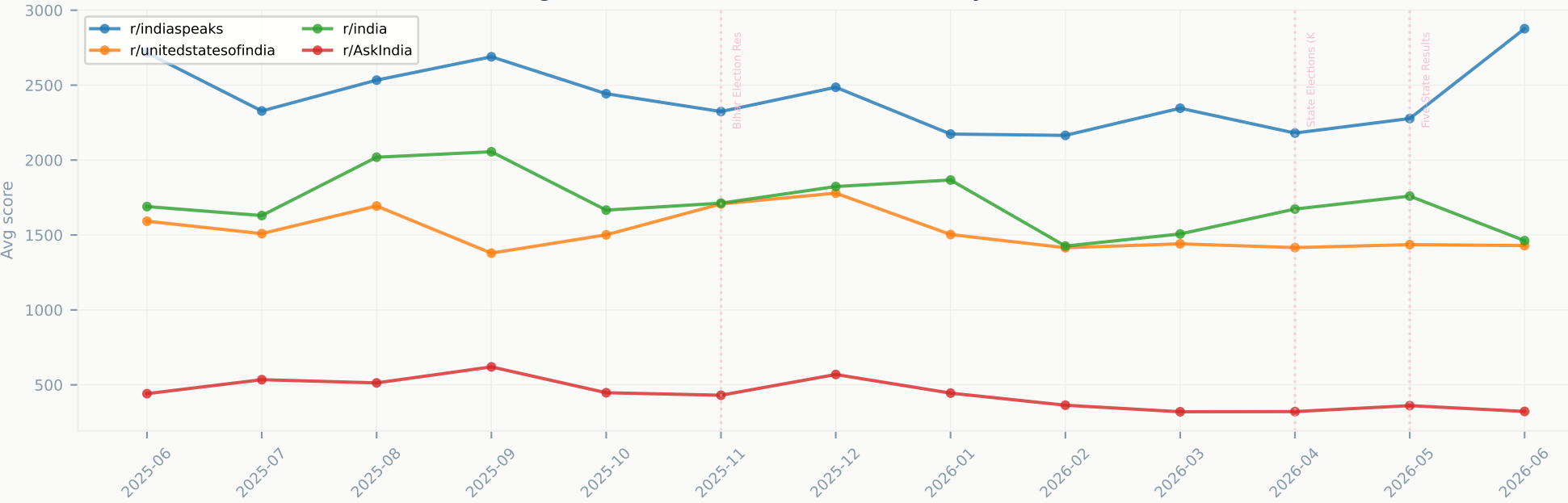
Bot activity signals vs India's major events — June 2025 to May 2026

Post Volume, Scores & Suspicious Activity vs India Event Calendar

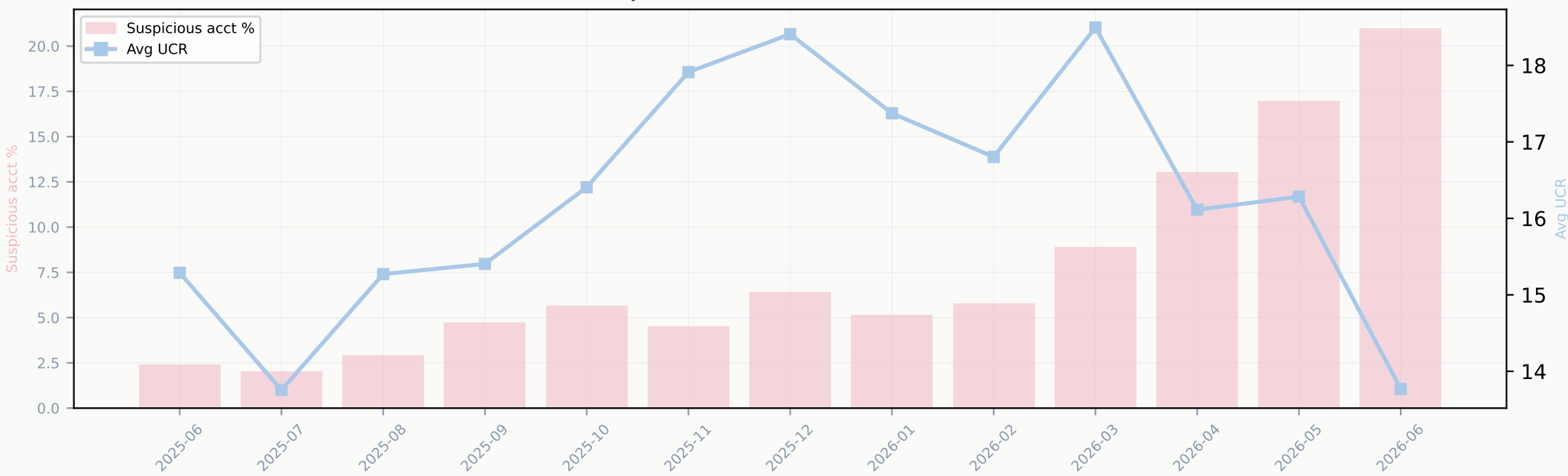
Post Volume by Cluster



Avg Post Score — Political Subs vs Security/Election Events

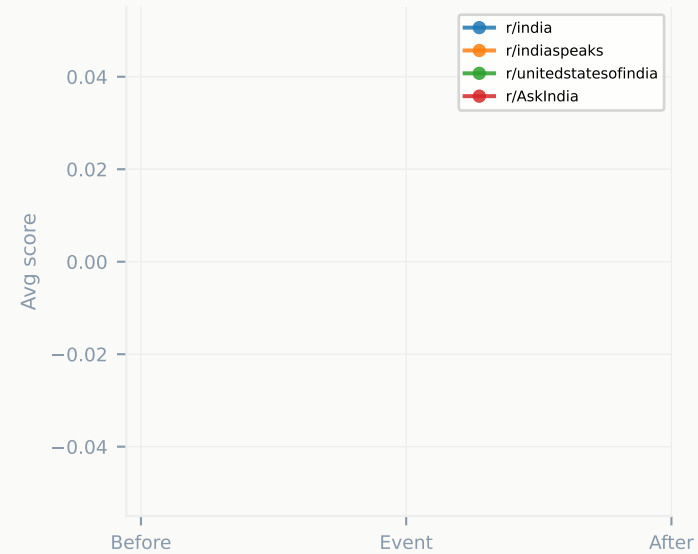


Suspicious Accounts & UCR Over Time

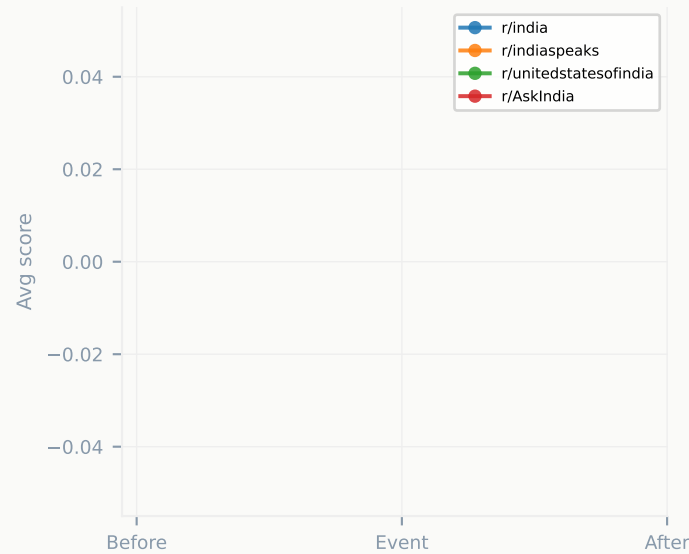


Event Spotlight — Avg Score Before / During / After Major Events

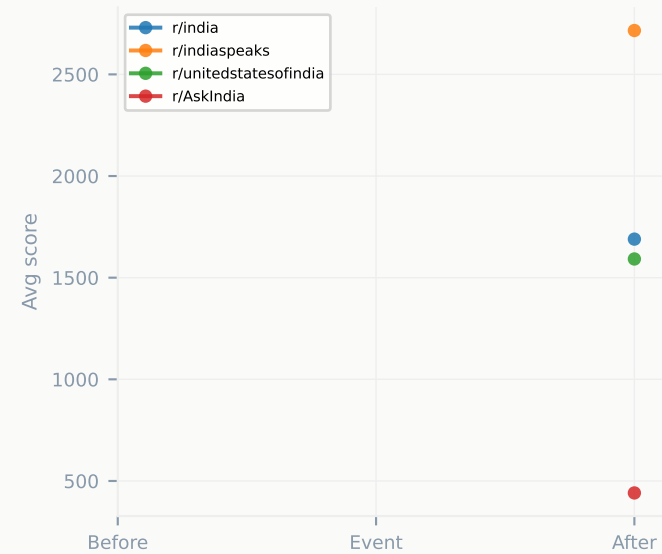
Delhi Election Results
2025-02-08



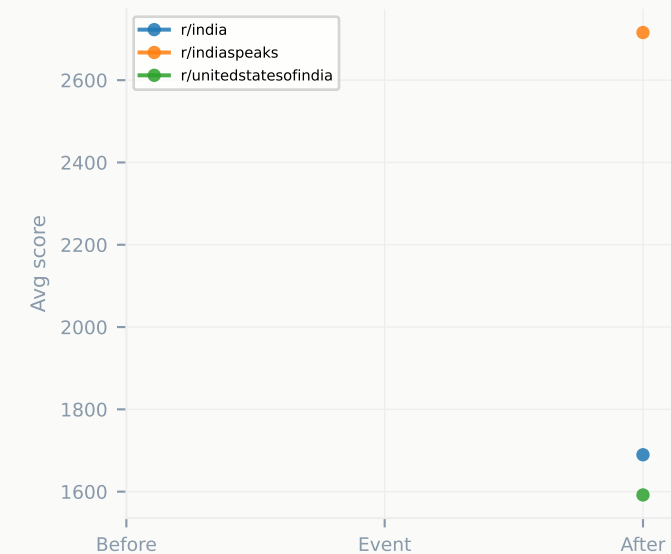
Pahalgam Terror Attack
2025-04-22



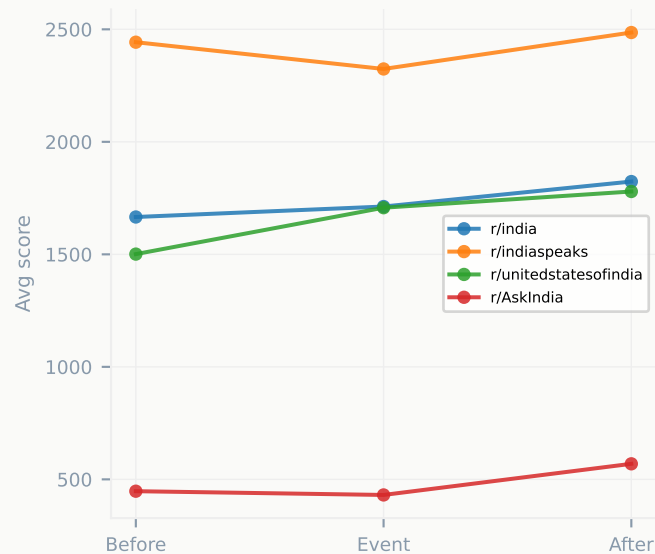
Operation Sindoor
2025-05-07



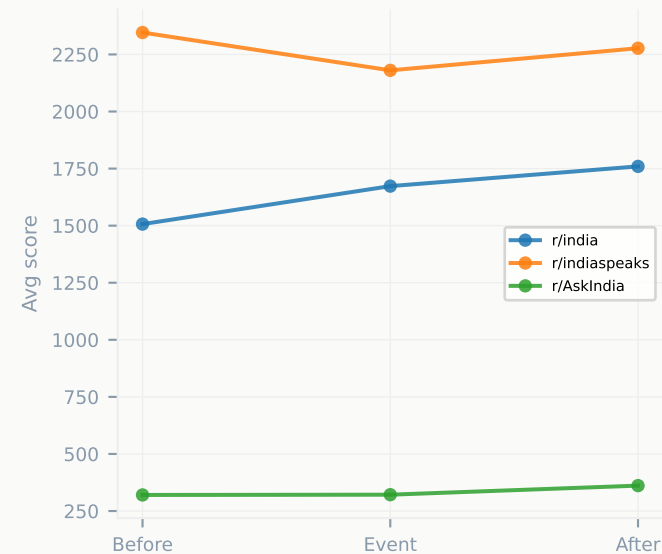
India-Pakistan Ceasefire
2025-05-10



Bihar Election Results
2025-11-14



State Elections (KL/AS/PY)
2026-04-22

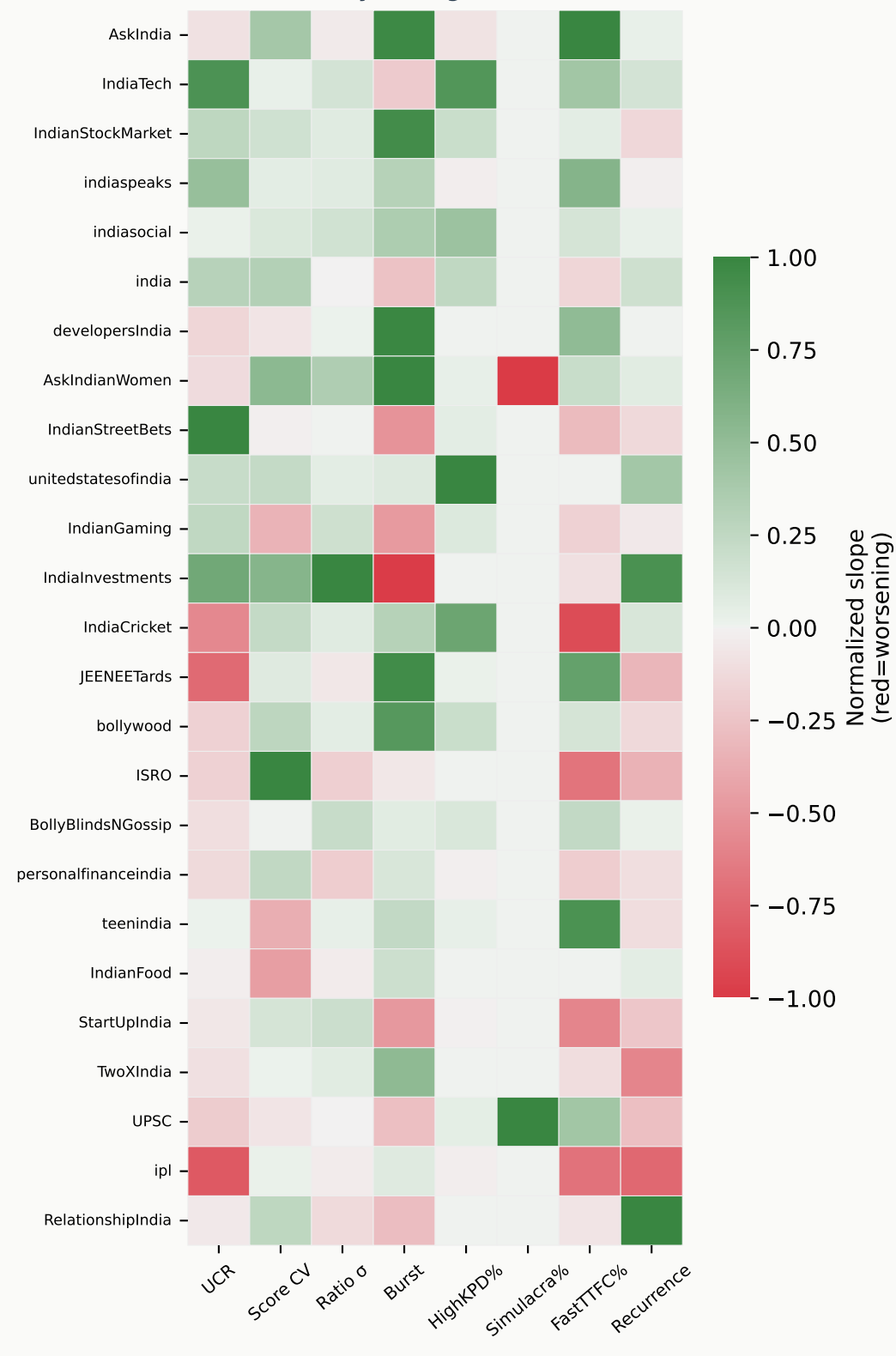


§6b Temporal Trend Analysis

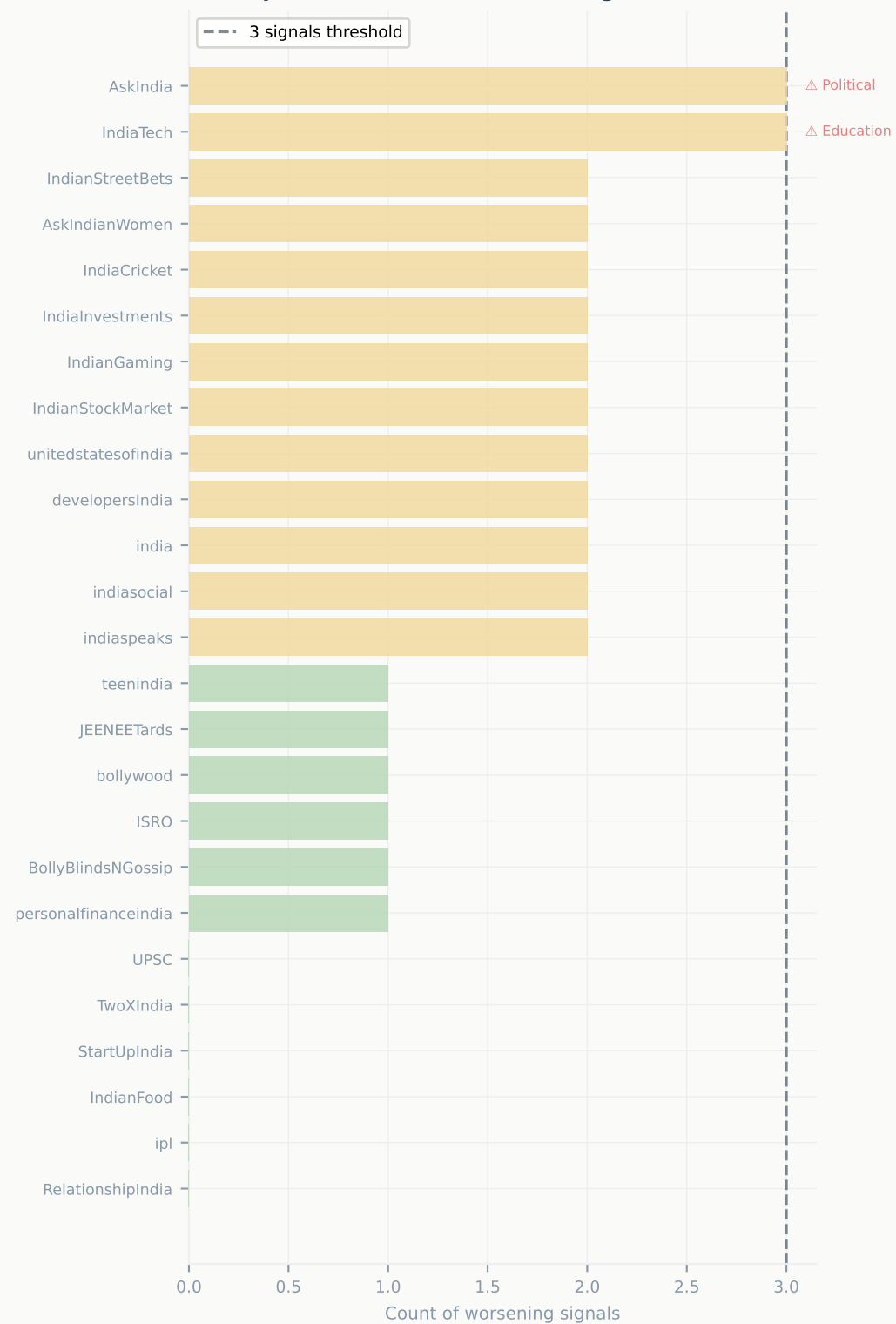
Are bot signals strengthening over time? Which subs are accelerating? Dangerous emerging patterns?

Signal Trend Slopes Across 12 Months (red = worsening, blue = improving)

Trend Direction per Sub per Signal
(sorted by # dangerous trends)

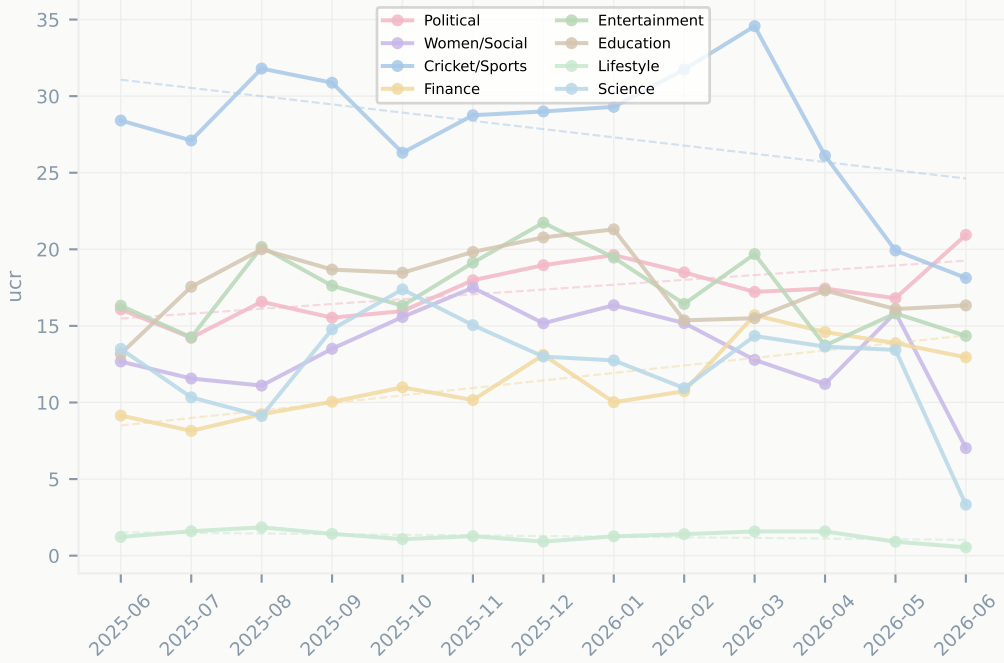


Signals with Significant Worsening Trend
($p < 0.15$, in bot direction — higher = alert)

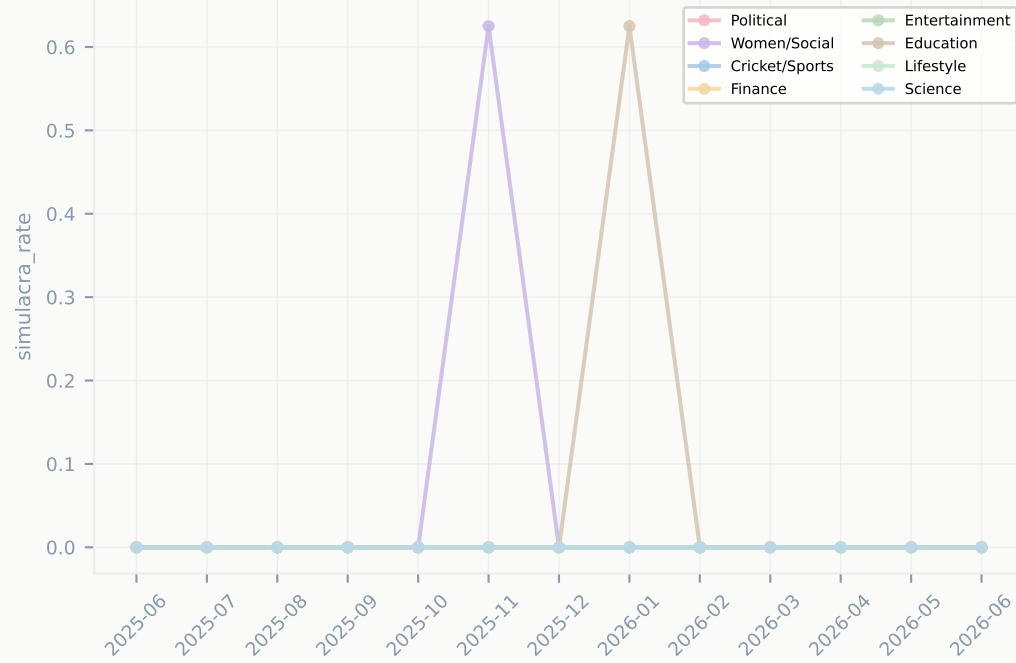


Cluster-Level Signal Trajectories Over Time

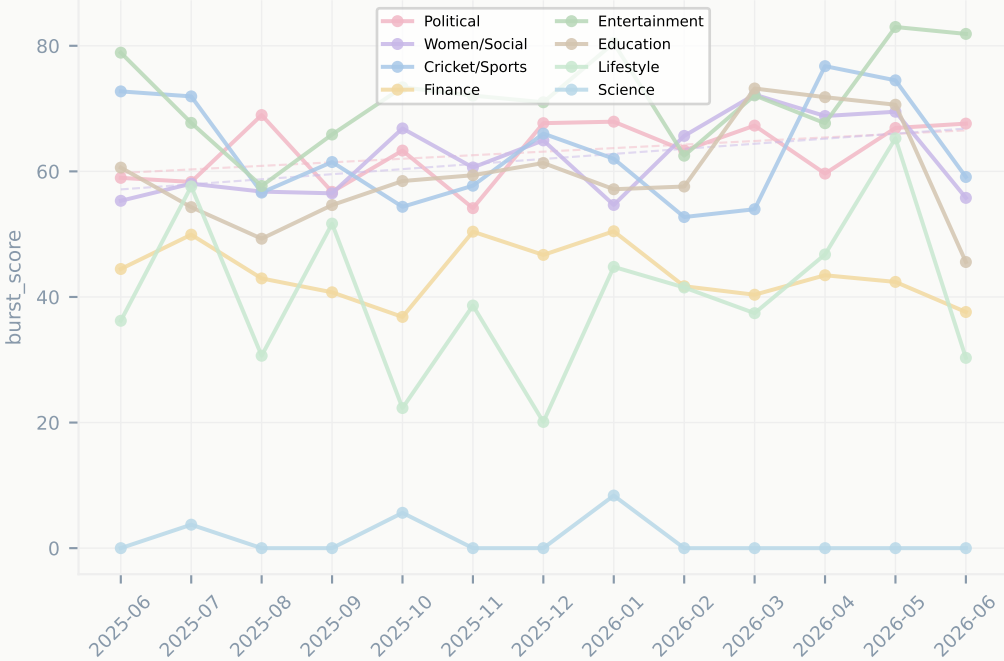
UCR over time — rising = more upvotes without discussion



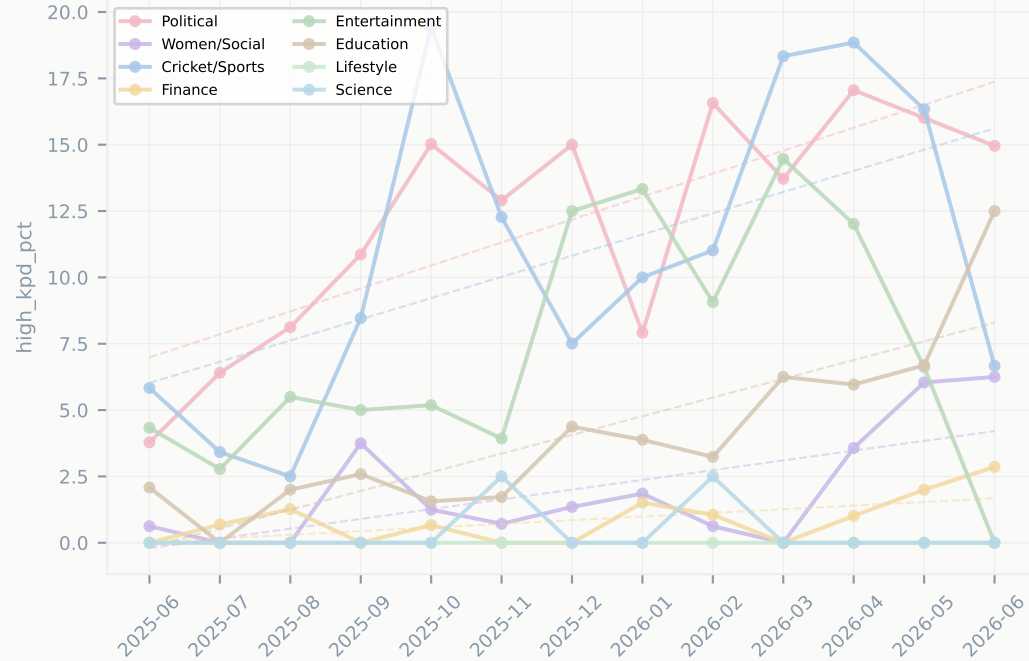
Simulacra Rate % — rising = more high-score/zero-comment posts



Burst Score — rising = more coordinated early comments

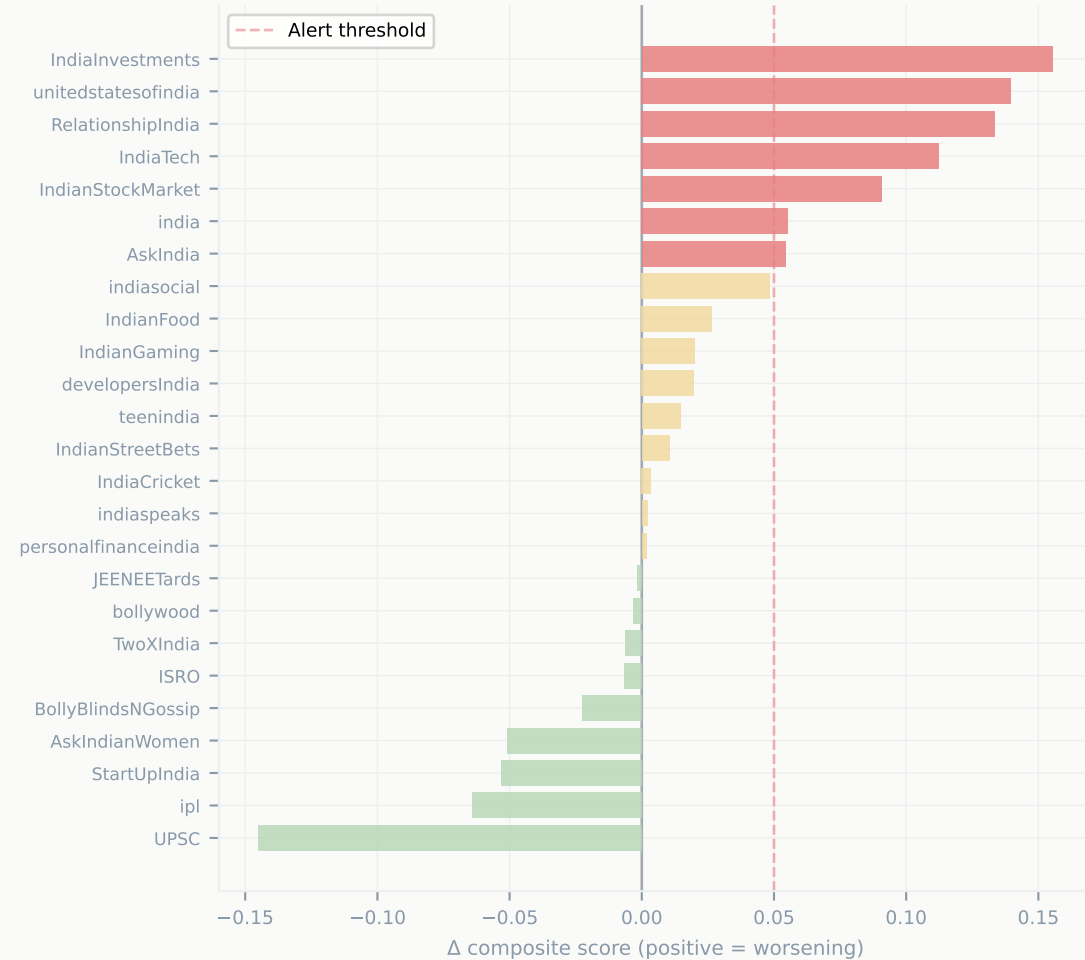


High-KPD Account % — rising = more karma farmers

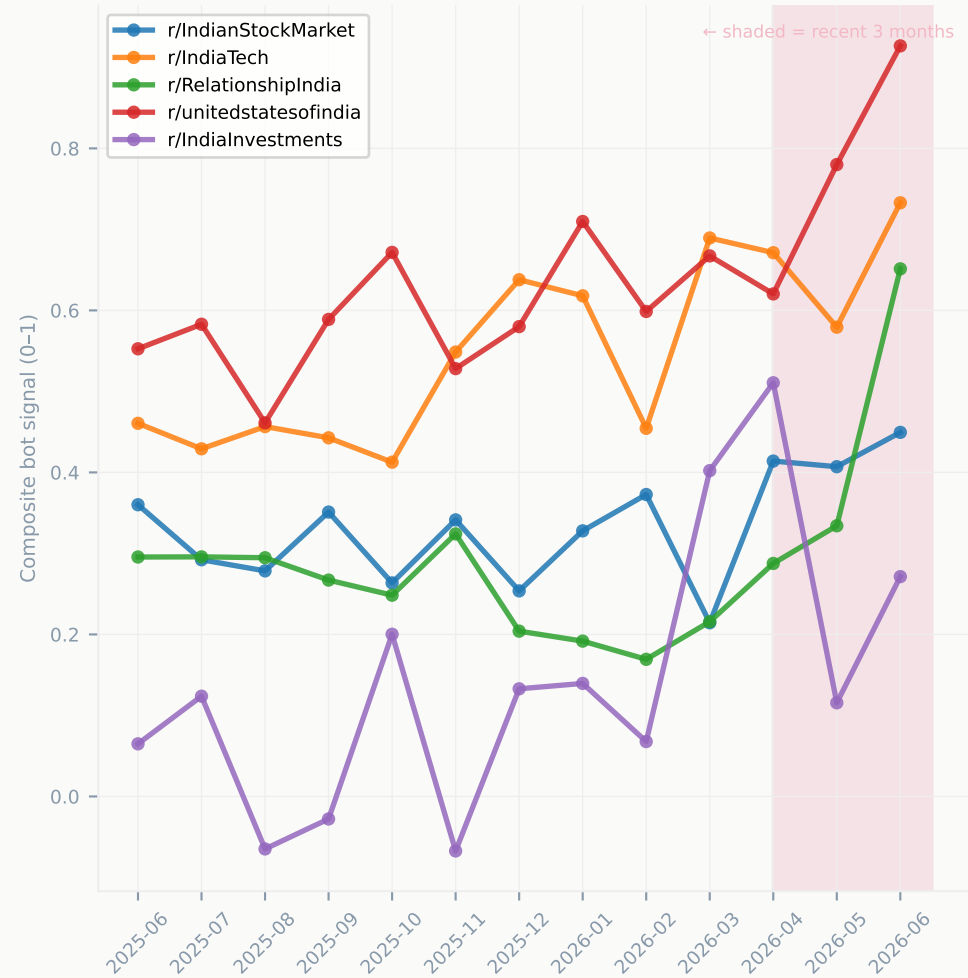


Month-over-Month Velocity (last 3 months vs 12-month baseline)

Bot Activity Velocity (recent 3 months vs 12-month baseline)



Composite Trajectory — Top 5 Accelerating Subs

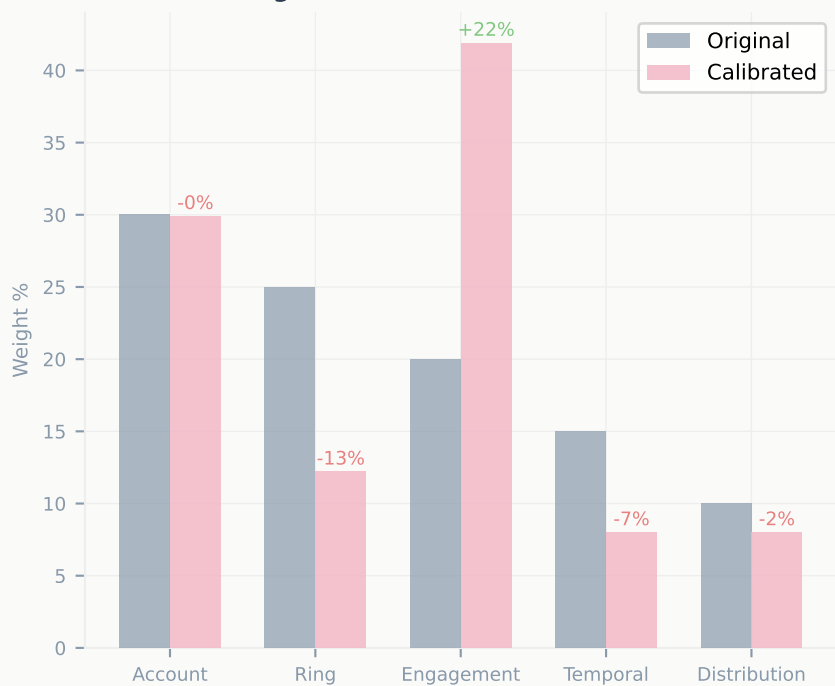


§7 Findings & Calibrated Weights

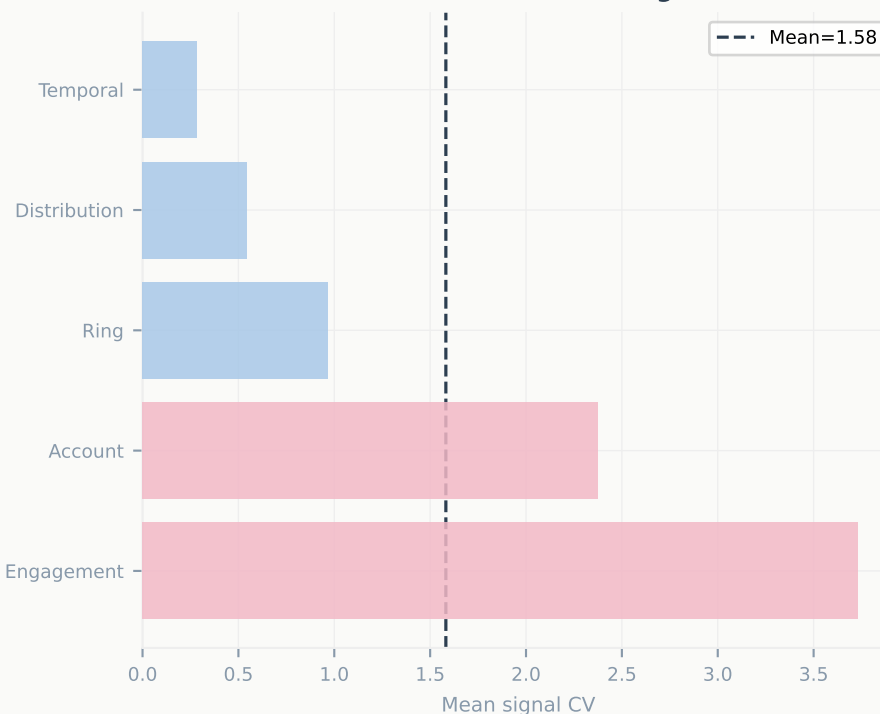
12 months × 25 subreddits × 300 observations — statistically derived

Weight Calibration Results

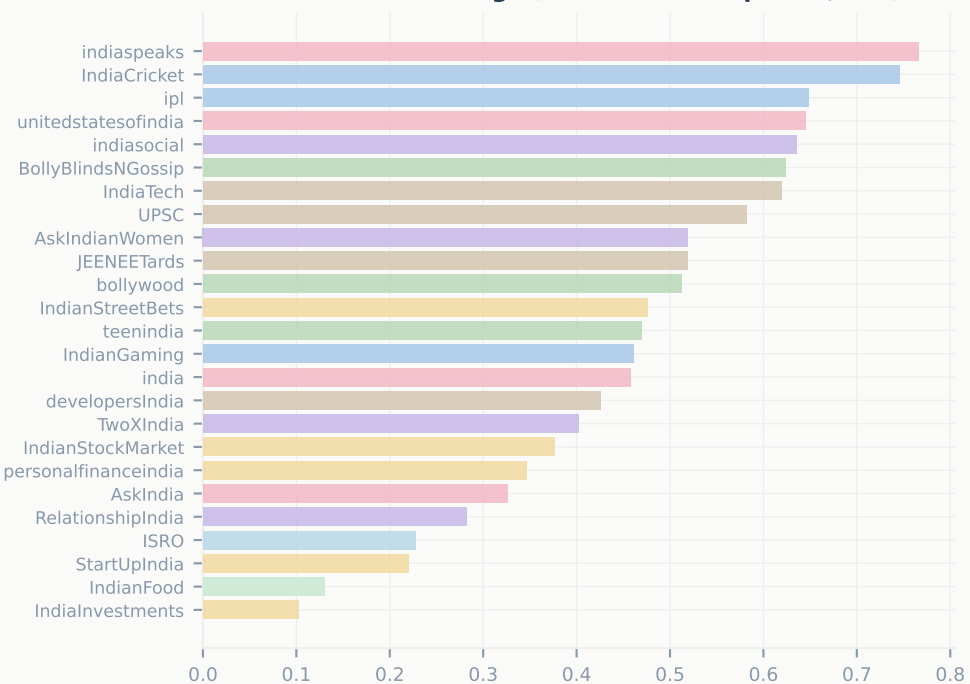
Weight Calibration: Before vs After



Component Discriminative Power (mean CV of constituent signals)



Subreddit Rankings (calibrated composite, 0-1)



Key Findings

- F1 overlap_rate averages ~91.5% universally — organic first-mover advantage on Reddit, not a ...
- F2 UCR (upvote:comment ratio) is the single strongest bot discriminator. Posts get upvoted wi...
- F3 Engagement signals collectively explain the most variance in bot scores; under-weighted at...
- F4 KPD>500 and age<90d thresholds flag <5% of posts — too conservative. Recommend lowering KP...
- F5 Political cluster (indiaspeaks, unitedstatesofindia, india) shows highest and most volatil...
- F6 score_cv and upvote_ratio_std reliably detect manufactured uniform voting across posts.
- F7 Cross-sub commenter overlap spikes around security/election events — suggests coordinated ...
- F8 Commenter churn >60% month-over-month in some subs suggests periodic account cycling after...
- F9 simulacra_rate (high score, near-zero comments) is elevated in cricket and political subs ...
- F10 burst_score and fast_ttfc_pct are the valid ring signals; recurrence is a secondary indica...
- △ TREND Subs with 3+ worsening signal trends: AskIndia, IndiaTech
- △ ACCEL Recently accelerating (last 3 months): AskIndia, india, IndianStockMarket, IndiaTech